

PdfPecker manual

Fast HTML and Markdown to PDF for .NET, with no dependencies.

For PdfPecker 1.9.0 · pdfpecker.net/docs

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Getting started

New and changed in PdfPecker 1.9.0 (since 1.8.0):

- HTML and CSS: CSS Changed
- HTML and CSS: CSS properties Changed
- HTML and CSS: Units and colours Changed
- HTML and CSS: Container queries New
- Paged media: Page rules and margin boxes Changed
- Paged media: Footnotes New
- Right-to-left, CJK and hyphenation: Chinese, Japanese and Korean line breaking New
- Fillable forms: Filling and flattening existing PDFs New
- Digital signatures: External signers: HSMs, smart cards and key vaults New
- Digital signatures: Verifying signatures New
- Digital signatures: Long-term validation New
- Text extraction New
- Limitations: Text and fonts Changed
- Limitations: CSS Changed
- Limitations: Reading and changing existing PDFs New

Every release in the changelog

PdfPecker turns HTML and Markdown into PDF, and PDF pages into PNG, JPEG or WebP images, in .NET. It has its own HTML parser, CSS engine, layout engine, PDF writer and PDF renderer, so it never starts a browser, ships no native binaries and depends on nothing but the .NET base class library. These pages describe PdfPecker 1.9.

```
using PdfPecker;

byte[] pdf = HtmlToPdf.Convert("<h1>Hello</h1><p>World</p>");
File.WriteAllBytes("hello.pdf", pdf);
```

Install

```
dotnet add package PdfPecker
```

The package targets `net8.0`, `net9.0` and `net10.0`, so it works on .NET 8 and later.

Usage

```
// String to bytes
byte[] pdf = HtmlToPdf.Convert(html);

// String to a stream
using (var file = File.Create("out.pdf")) HtmlToPdf.Convert(html, file);

// ASP.NET Core
app.MapGet("/invoice", () => Results.File(HtmlToPdf.Convert(html), "application/pdf",
"invoice.pdf"));

// File to file; relative <img> and <link> paths resolve against the HTML file's folder
HtmlToPdf.ConvertFile("invoice.html", "invoice.pdf");

// Off the calling thread
byte[] pdf = await HtmlToPdf.ConvertAsync(html);
```

All methods are static and thread-safe, so you can convert many documents in parallel. Page size, margins, headers, footers and the rest are set with `PdfOptions`.

System requirements

Required: the .NET 8, .NET 9 or .NET 10 runtime (or later). Nothing else: the library is pure managed code with no native binaries, browser or third-party packages, so it runs wherever .NET runs (Windows, Linux, macOS; x64 and ARM64). The rendered pages are the same on every version; only the compressed bytes can differ, because each .NET version ships its own zlib build.

Recommended:

- Register the fonts your templates use with `options.Fonts` (see Fonts); other fonts fall back to Helvetica, Times or Courier metrics.
- Memory and CPU are small. Measured on a 4-core 2.1 GHz Xeon cloud VM (Linux, .NET 10, Release), each document in a fresh process:

Document	Pages	Time, warm	Time, first in process	Peak process memory	Smallest heap that works
Payslip, invoice, voucher	1 to 3	1 to 3 ms	0.2 s	50 MB	under 16 MB
Same with PDF/A or embedded fonts	1 to 3	3 to 5 ms	0.2 s	56 MB	under 16 MB
Report with tables	30	30 to 40 ms	0.3 s	75 MB	under 16 MB
Report with tables	1,000	1.7 s	3 s	480 MB	245 MB

  The process itself starts at about 32 MB; a conversion uses about one core, so throughput grows with cores. PDF/A and font embedding add a few milliseconds and about 5 MB for the embedded font data. Memory grows roughly with page count, so plan about 0.5 MB per page for very long documents.

From a clone of the PdfPecker repository, `dotnet run --project bench/PdfPecker.Benchmarks -c Release -- --requirements [your.html ...]` measures this on your own machine and documents.

Where to go next

- Options: page size, margins, headers and footers, document properties and file access.
- HTML and CSS support and Paged media: what your templates can use.
- Fonts: register your own fonts and use any Unicode text.
- Limitations: what PdfPecker does not do (yet).

HTML to PDF in C#

This guide shows how to convert HTML to PDF in C# with PdfPecker, from a one-line conversion to an invoice built from a template and served from ASP.NET Core. PdfPecker is a pure C# library: it does not start Chrome or another browser, needs no native binaries such as wkhtmltopdf, and has no NuGet dependencies, so the same code runs on Windows, Linux, macOS, in Docker and in serverless functions.

Install the NuGet package

```
dotnet add package PdfPecker
```

PdfPecker works with .NET 8, .NET 9 and .NET 10 and is free under the MIT license.

Convert an HTML string to PDF

```
using PdfPecker;

byte[] pdf = HtmlToPdf.Convert("<h1>Hello</h1><p>World</p>");
File.WriteAllBytes("hello.pdf", pdf);
```

`HtmlToPdf.Convert` returns the PDF as a byte array. To write it straight to a file, a `MemoryStream` or an HTTP response, pass a stream instead:

```
using (var file = File.Create("hello.pdf")) HtmlToPdf.Convert(html, file);
```

Convert an HTML file to PDF

```
HtmlToPdf.ConvertFile("invoice.html", "invoice.pdf");
```

Relative image and stylesheet paths in the file, such as `<img src="logo.png">` or `<link rel="stylesheet" href="invoice.css">`, are read from the HTML file's folder. When you convert a string, set `BaseDirectory` to say where those files are:

```
var options = new PdfOptions { BaseDirectory = "/var/app/templates" };
byte[] pdf = HtmlToPdf.Convert(html, options);
```

Generate an invoice from a template

Build the HTML with whatever produces a string, such as a Razor view rendered to a string, a template engine, or plain string interpolation, and convert the result. Encode values that come from users or a

database, so that a customer name with a `<` in it cannot break the page.

```
using System.Net;
using PdfPecker;

record Line(string Description, int Quantity, decimal Price);

static byte[] Invoice(string customer, IEnumerable<Line> lines)
{
    var rows = string.Concat(lines.Select(l =>
$"<tr><td>{WebUtility.HtmlEncode(l.Description)}</td><td>{l.Quantity}</td><td>{l.Quantity *
l.Price:N2}</td></tr>"));
    var total = lines.Sum(l => l.Quantity * l.Price);

    var html = $$"""
        <style>
          body { font-family: Helvetica, sans-serif; font-size: 10pt }
          table { width: 100%; border-collapse: collapse }
          th, td { padding: 6px; border-bottom: 1px solid #ddd; text-align: left }
          td:last-child, th:last-child { text-align: right }
        </style>
        <h1>Invoice</h1>
        <p>To: {{WebUtility.HtmlEncode(customer)}}</p>
        <table>
          <thead><tr><th>Item</th><th>Qty</th><th>Amount</th></tr></thead>
          <tbody>{{rows}}</tbody>
        </table>
        <p style="text-align: right"><b>Total: {{total:N2}}</b></p>
        """;

    return HtmlToPdf.Convert(html, new PdfOptions { Title = "Invoice" });
}
```

Table headers repeat on every page when a table runs over a page break, and flexbox, grid and the rest of the CSS PdfPecker supports work in templates as they do in a browser.

Return a PDF from ASP.NET Core

```
app.MapGet("/invoice/{id}", (int id) =>
{
    byte[] pdf = Invoice("ACME Ltd", LoadLines(id));
    return Results.File(pdf, "application/pdf", $"invoice-{id}.pdf");
});
```

In an MVC controller, `return File(pdf, "application/pdf", "invoice.pdf");` does the same. All PdfPecker methods are static and thread-safe, so many requests can convert at once, and `HtmlToPdf.ConvertAsync` moves the work off the calling thread.

Page size, margins, headers and footers

```
var options = new PdfOptions
{
    PageSize = PageSize.Letter,                          // A4 is the default
    Margins = PageMargins.FromMillimeters(20, 15, 20, 15),
    HeaderText = "ACME Ltd",
    FooterText = "Page {page} of {pages}",
};
byte[] pdf = HtmlToPdf.Convert(html, options);
```

The page can also be set from CSS, which keeps it with the template:

```
@page {
  size: A4 landscape;
  margin: 20mm 15mm;
  @bottom-right { content: "Page " counter(page) " of " counter(pages) }
}
```

For a header or footer written in HTML, with a logo or a table, see Paged media, and for every setting see Options.

Fonts

Text without a registered font uses the standard PDF fonts Helvetica, Times and Courier. To use your own font, register the TrueType, OpenType, WOFF or WOFF2 file and refer to it by its family name in CSS; only the characters the document uses are embedded.

```
var options = new PdfOptions();
options.Fonts.AddFile("fonts/OpenSans-Regular.ttf");
byte[] pdf = HtmlToPdf.Convert("<p style=\"font-family: 'Open Sans'\">Hello</p>", options);
```

Fonts covers `@font-face`, bold and italic files, and what happens when a font is missing.

Images

`<img>` and CSS backgrounds take PNG, JPEG, GIF, WebP and SVG files, and `data:` URLs. Files are read only from inside `BaseDirectory`, and PdfPecker never downloads anything by itself: to load `http` or `https` images, hand them over with `ResourceResolver`.

```
var http = new HttpClient();
var options = new PdfOptions
{
    ResourceResolver = url => url.StartsWith("https://cdn.example.com/")
        ? http.GetByteArrayAsync(url).GetAwaiter().GetResult()
        : null,
};
```

Common questions

Does PdfPecker need Chrome, Puppeteer or wkhtmltopdf? No. It has its own HTML parser, CSS engine, layout engine and PDF writer, written in C#. There is nothing to install on the server besides the .NET runtime.

Does it run on Linux, in Docker and in Azure Functions or AWS Lambda? Yes. It is pure managed code, so it runs wherever .NET 8 or later runs, on x64 and ARM64.

Does it run JavaScript? No. PdfPecker converts the HTML and CSS you give it, which suits templates you control, such as invoices, reports, payslips and letters. Pages that build themselves with JavaScript need a browser-based converter; Compare lists them.

How fast is it? A one-page invoice takes about a millisecond once the process is warm. See Performance for the measurements.

Is it free for commercial use? Yes. PdfPecker is open source under the MIT license.

Can I also convert Markdown, or make PDF/A? Yes: `MarkdownToPdf.Convert` takes Markdown (see Markdown), and `PdfOptions.Conformance` makes PDF/A for archiving (see PDF/A, passwords and watermarks).

Options

`PdfOptions` controls the page, the document properties and how resources are loaded. Every conversion method takes one; leave it out for A4 pages with the defaults below.

```
var options = new PdfOptions
{
    PageSize = PageSize.Letter.Landscape,            // A3, A4 (default), A5, Letter, Legal or
new PageSize(w, h) in points
    Margins = PageMargins.FromMillimeters(20, 15, 20, 15),
    Scale = 1,                                       // like Chrome's print scale (see its
docs to match IronPDF's output)
    TruncateBandsToWholePoints = false,              // round header/footer height attributes
down to whole points, as Chrome does
    BaseDirectory = "/var/app/templates",            // where relative image/stylesheet paths
are resolved
    ResourceResolver = url => myCache.Get(url),      // serve http(s) or embedded resources
yourself; null = default handling
    UserStyleSheet = "body { font-size: 10pt }",     // extra CSS applied after the
document's own
    FooterText = "Page {page} of {pages}",           // also HeaderText
    Title = "Invoice 42", Author = "Accounts",       // document properties (Title defaults
to <title>)
    Keywords = "invoice", Creator = "MyApp",         // also Subject, and Producer (default
"PdfPecker")
    CustomProperties = { ["RunId"] = "42" },         // your own properties, in the PDF's
document information
    Bookmarks = true,                                // PDF bookmarks from h1-h6 (default)
    FormFields = true,                               // inputs, selects and textareas become
fillable fields (default)
    Compress = true,                                 // Flate-compress page content (default)
    AllowFileAccess = false,                         // true: also read files outside
BaseDirectory (trusted HTML only)
    MaxPages = 1000,                                 // refuse longer documents with
PdfLimitException (default 100,000)
};
byte[] pdf = HtmlToPdf.Convert(html, options);
```

Files, network and untrusted HTML

Local images and stylesheets are read only from inside `BaseDirectory` (the input file's folder for `ConvertFile`): absolute paths, `file:` URLs, `..` and symbolic links that lead outside it are ignored unless `AllowFileAccess = true`, and converting a string without a `BaseDirectory` reads no files at all. For HTML from untrusted users, keep it that way, set a low `MaxPages`, and pass a cancellation token with a timeout (`HtmlToPdf.Convert(html, options, token)`): cancelling stops the conversion where it is.

Page size from CSS

A CSS `@page { size: A4 landscape; margin: 15mm }` rule overrides the page size and margins unless `IgnoreCssPageRules` is set. PdfPecker never makes network requests on its own: `http(s)` images and stylesheets are only loaded through `ResourceResolver`.

Document properties

Custom properties are written to the document information dictionary, where Acrobat shows them under Document Properties > Custom and `pdfinfo -custom` lists them. For PDF/A they are also written to the XMP metadata (Acrobat's `pdfx` namespace, with the extension schema PDF/A requires), alongside Keywords, Creator and Producer.

More options

The other options have pages of their own: `Fonts`, `EmbedStandardFonts`, `MissingFont` and `MissingCharacter`, `Language` and `Hyphenation`, `Watermark`, `Security` and `Conformance`, `Accessibility`, `EInvoice`, `Attachments`, `Signature` and `FormFields`.

HTML and CSS support

> **Changed in 1.9.0:** CSS, CSS properties, Units and colours, Container queries.

PdfPecker lays out HTML and CSS the way Chrome prints it. Anything not listed here should be assumed missing; the known gaps are listed under Limitations.

HTML

The common document elements: headings, paragraphs, `div` / `section`-style blocks, `span`, `a` (external links and `#id` links inside the document), `b` / `strong`, `i` / `em`, `u`, `s`, `sub`, `sup`, `small`, `big`, `mark`, `code`, `pre`, `blockquote`, `hr`, `br`, `ul` / `ol` / `li` (nested, with `start` and `value`), `dl`, `table` (with `thead`, `tbody`, `tfoot`, `caption`, `colspan`, `rowspan`, `col width`), `img`, inline `<svg>`, and legacy attributes such as `align`, `valign`, `bgcolor`, `width`, `height`, `border`, `cellpadding`, `cellspacing` and `<font>`. The `dir` attribute (`ltr`, `rtl` and `auto`), `<bdi>` and `<bdo>` set the text direction. The parser is forgiving about unclosed and misnested tags, the way browsers are.

Markdown is converted to this HTML first; see Markdown.

CSS Changed in 1.9.0

`<style>` blocks, `<link rel="stylesheet">`, `@import` (with `layer()`, `supports()` and media queries; a relative URL is relative to the importing stylesheet), inline `style` attributes, `@media` (media queries level 4: `print`, `width` / `height` and `orientation` of the page area, ranges such as `(400px <= width <= 800px)`, `and`, `or`, `not`), `@supports` (property tests, `selector()`, `and` / `or` / `not`), `@layer` cascade layers, CSS nesting (`&`), `@property` (initial values and inheritance of registered custom properties), `@container` (see Container queries) and `@page` (see Paged media). Selectors: type, `.class`, `#id`, `*`, attribute selectors (`[a]`, `=`, `~=`, `|=`, `^=`, `$=`, `*=`, with `i` / `s` flags), descendant, `>`, `+`, `~`, `:first-child`, `:last-child`, `:only-child`, `:nth-child()` and `:nth-last-child()` (incl. `of S`), `:first-of-type`, `:last-of-type`, `:only-of-type`, `:nth-of-type()`, `:nth-last-of-type()`, `:not()`, `:is()`, `:where()` and `:has()` (all with lists of complex selectors, such as `:is(nav > a, .menu li)`, nested to any sensible depth; `:is()` and `:where()` skip entries they do not understand), `:root`, `:link`, `:empty`, `:lang()`, form states (`:checked`, `:disabled`, `:enabled`, `:required`, `:optional`, `:read-only`, `:placeholder-shown` and the like), `::before`, `::after`, `::marker`, `::first-letter` (drop caps, floated or not) and `::first-line`, with specificity and `!important` handled as in browsers. As when Chrome prints, interaction states such as `:hover` and `:focus` never match.

CSS properties Changed in 1.9.0

Area	Properties

Area	Properties
Box model	`display` (block, inline, inline-block, flex, inline-flex, grid, inline-grid, list-item, table parts, none), `margin`, `padding` (also the logical `margin-inline` / `-block` and `padding-inline` / `-block` properties and their `-start` / `-end` longhands), `border` (solid, dashed, dotted, double, groove, ridge, inset, outset), `border-image` (images, SVG files and gradients, with `-slice` incl. `fill`, `-width`, `-outset` and `-repeat`: stretch, repeat, round, space), `border-radius` (incl. elliptical `/` radii and percentages; rounds backgrounds, borders, shadows and images), `box-shadow` (outer and `inset` shadows; a blur is drawn as fading bands), `overflow` (`hidden`, `clip`, `auto` and `scroll` clip to the padding box, following its rounded corners, also in table cells; `overflow-x` / `overflow-y`), `outline` and `outline-offset` (all border styles; follows rounded corners), `width`, `height`, `min-` / `max-width`, `min-height`, `box-sizing`, `aspect-ratio` (incl. `auto <ratio>`; also in shrink-to-fit widths, with `min-` / `max-height`, and on tables), `object-fit` and `object-position` on images (SVG files too), `clip-path` (`inset()`, `circle()`, `ellipse()`, `polygon()`, `path()`, reference boxes, and `url(#id)` of a `clipPath` in inline SVG)
Text	`color`, `font`, `font-family`, `font-size`, `font-weight`, `font-style`, `font-stretch`, `font-variation-settings`, `font-optical-sizing` (see Variable fonts), `line-height`, `text-align` (incl. justify, `start` and `end`), `direction`, `unicode-bidi`, `text-decoration`, `text-transform`, `text-indent`, `letter-spacing`, `word-spacing`, `white-space`, `overflow-wrap` / `word-wrap`, `word-break` (`break-all`, `keep-all`), `line-break` (`auto`, `strict`, `normal`, `loose`, `anywhere`; see CJK line breaking), `vertical-align`, `text-shadow` (several shadows, a true Gaussian blur, and underlines and line-throughs get the shadow too), `text-overflow` (`ellipsis`, `clip` or a string such as `" [more]"`, on a box that clips its overflow), `text-decoration-style` (`solid`, `double`, `dotted`, `dashed`, `wavy`), `text-decoration-color`, `text-decoration-thickness`, `text-underline-offset`, `text-align-last`, `tab-size`, `font-variant: small-caps` (synthesized from capitals when the font has none), `font-variant-numeric` and `font-feature-settings` (see OpenType features), `line-clamp` and `-webkit-line-clamp` (with an ellipsis), `hyphens: manual`
Backgrounds	`background-color`, `background-image` and `background` with `url()` images (PNG, JPEG, GIF, WebP, BMP, TIFF, and SVG files drawn as vectors), `linear-gradient()`, `radial-gradient()`, `conic-gradient()` (pie charts and colour wheels, drawn as smooth PDF shadings) and their `repeating-` forms (several layers allowed); `background-repeat` (incl. `space`, `round`), `background-position` (incl. edge offsets like `right 10px bottom 5px`), `background-size` (`cover`, `contain`, lengths), `background-origin`, `background-clip`, `background-blend-mode`. Repeated images are written once as PDF tiling patterns. `url()` in a linked stylesheet is relative to the stylesheet and follows the same file access rules as `<img>`. As in Chrome's print output, the background of `<html>`, or else of `<body>`, fills every page area (images and gradients too, positioned against the root box and continuing from page to page); `@page { background: <colour> }` also colours the page margins. `opacity` on SVG
Flexbox	`flex-direction`, `flex-wrap`, `flex-flow`, `flex`, `flex-grow` / `-shrink` / `-basis`, `order`, `justify-content`, `align-items`, `align-self`, `align-content`, `gap`, `margin: auto`
Grid	`grid-auto-flow: dense` packing, `grid-template-columns` / `-rows` (lengths, `%`, `fr`, `auto`, `min-content`, `max-content`, `minmax()`, `fit-content()`, `repeat()` incl. `auto-fill` / `auto-fit`), `grid-auto-rows` / `-columns`, `grid-auto-flow`, `grid-template-areas`, `grid-template`, named lines (`[name]`), `grid-column` / `-row` (line numbers, names and `span`), `grid-area` (incl. area names), `gap`, `justify-items` / `-self`, `align-items` / `-self`, `justify-content`, `align-content`, `place-*`
Columns	`column-count`, `column-width`, `columns`, `column-gap`, `column-rule` (all border styles), `column-span: all`
Variables	custom properties (`--name: value`) and `var(--name, fallback)`, anywhere including shorthands
Math functions	`calc()`, `min()`, `max()`, `clamp()`, `round()`, `mod()`, `rem()`, `abs()`, `sign()`, `sin()`, `cos()`, `tan()`, `asin()`, `acos()`, `atan()`, `atan2()`, `pow()`, `sqrt()`, `hypot()`, `log()` and `exp()` for lengths, numbers and angles, nested in any combination and mixing units (`calc(100% - 2em)`, `min(100%, 600px)`, `clamp(1rem, 2.5vw, 2rem)`), with `var()` inside them. Percentages resolve against the property's own reference (the containing block's width for `width` and margins, the parent's font size for `font-size`), so they work in widths, heights, margins, padding, flex bases, grid tracks, gaps, positions, translations, font sizes and borders

Area	Properties
Positioning	`float` and `clear`; `position` (`relative`, `absolute`, `fixed`; `sticky` is drawn as `relative`, as in Chrome's print output) with `top`, `right`, `bottom` and `left`; `z-index`
Transparency	`opacity` (a block is painted as one PDF transparency group; an inline `<span>` fades its text and background), translucent colours everywhere (`rgba()`, `hsla()`, `#rrggbbaa`, `transparent`), gradients that fade to `transparent` (as a soft mask). PDF/A-2 and 3 allow all of this
Effects	`filter`: `grayscale()`, `sepia()`, `saturate()`, `hue-rotate()`, `invert()`, `opacity()`, `brightness()` and `contrast()` recolour the box's text, borders and backgrounds as vectors (images through their pixels), while `blur()` and `drop-shadow()` are applied to the rendered box at 144 dpi, and text under `blur()` stays selectable and searchable; `mix-blend-mode` (all separable and non-separable modes, as PDF blend modes) and `isolation`
Transforms	`transform` (`translate()`, `translateX()`/`Y()`, `scale()`, `scaleX()`/`Y()`, `rotate()`, `skew()`, `skewX()`/`Y()`, `matrix()`), `transform-origin`, and the individual `translate`, `rotate` and `scale` properties
Generated content	`::before`, `::after` and `::marker` with `content` (strings, `attr()`, `counter()`, `counters()`, `open-quote`/`close-quote`, and for paged media `target-counter()`, `target-text()`, `leader()`, `string()` and `content()`), `counter-reset`, `counter-increment`, `counter-set`, `string-set`
SVG	`fill`, `stroke`, `stroke-width`, `fill-opacity`, `stroke-opacity`, `fill-rule`, `stroke-linecap`, `stroke-linejoin`, `stroke-miterlimit`, `clip-path`, `clip-rule`, `mask`, `mask-type`, `filter`, `marker`, `marker-start`/`-mid`/`-end`, `flood-color`, `flood-opacity`, `color-interpolation-filters`, `text-anchor`, `dominant-baseline` (as CSS or presentation attributes)
Lists	`list-style-type` (every predefined style of CSS Counter Styles 3, such as `lower-greek`, `hebrew`, `japanese-formal`, `cjk-decimal`, `disclosure-open`, written as Chrome writes them, or a string such as `'– '`), `@counter-style` rules (every system and descriptor) and `symbols()`, also in `counter()` and `counters()`; `list-style-image` (raster images and SVG files, falling back to the type when the image does not load), `list-style`, `::marker` (content, colour, font)
Tables	`border-collapse` (each shared border drawn once, the wider one winning), `border-spacing`, cell `vertical-align`, automatic column widths (a column width that mixes a percentage with a length, such as `calc(50% - 10px)`, counts as `auto`, as in Chrome), `table-layout: fixed`, a table `height` shared out over the rows
Paging	`page-break-before`/`after`/`inside`, `break-before`/`after`/`inside` (`always`, `page`, `avoid`), `orphans`, `widows`, `@page` size, margins and background, `@page :first`/`:left`/`:right`, the sixteen page-margin boxes (`@top-center` and the like), named pages (`page`), `string-set`, footnotes (`float: footnote`, `@footnote`) (see Paged media)

Units and colours Changed in 1.9.0

Units: `px`, `pt`, `em`, `rem`, `%`, `mm`, `cm`, `in`, `pc`, `ex`, `ch`, `vw`, `vh`, `vmin`, `vmax` (the viewport is the page's content area, minus any running header and footer, in whole pixels as in Chrome), and the container query units `cqw`, `cqh`, `cqi`, `cqb`, `cqmin` and `cqmax`. Colours: names, `#rgb`, `#rrggbb`, `#rrggbbaa`, `rgb()`, `rgba()`, `hsl()`, `hsla()` (comma or space syntax), `hwb()`, `lab()`, `lch()`, `oklab()`, `oklch()`, `color()` (sRGB, Display P3 and the other predefined spaces, mapped to sRGB), `color-mix()`, `light-dark()` (the light value), `currentColor`, and relative colours (`rgb(from var(--brand) r g b / 50%)`, `oklch(from teal calc(l + 0.1) c h)` and the like, in every colour function, with the channels usable in math functions).

Container queries New in 1.9.0

`container-type` (`inline-size` or `size`), `container-name` and the `container` shorthand make an element a query container, and `@container` rules style its descendants by the container's size: `width`, `height`, `inline-size`, `block-size`, `aspect-ratio` and `orientation`, with `min-`/`max-` prefixes or range syntax, combined with `and`, `or` and `not`. A name picks a container (`@container sidebar (width < 300px)`), and `style(--compact: 1)` queries a custom property. The `cq` units are percentages of the nearest container's size. One component can then lay itself out for a narrow sidebar and a wide main column without knowing where it is placed.

```
.card-slot { container: card / inline-size }
.card { display: grid; gap: 8px }
@container card (width >= 400px) { .card { grid-template-columns: 120px 1fr } }
.card h3 { font-size: clamp(14px, 5cqi, 22px) }
```

A document that uses container queries is laid out again until the container sizes settle, which takes about two to four times as long; documents without them are laid out as before.

Pagination

Lines, images and table rows are never cut in half at a page boundary; table header rows repeat on every page a table spans; backgrounds and borders continue across pages. Flex lines and grid rows move to the next page as a whole when they don't fit. A paragraph that splits leaves at least `orphans` lines (default 2) at the bottom of a page and carries at least `widows` lines (default 2) to the next. Headings keep with the content that follows them (`break-after: avoid` in the default styles), and `break-before`/`break-after: avoid` does the same for any block. `break-inside: avoid` keeps a block or a whole table on one page when it fits.

Links and bookmarks

`<a href>` becomes a clickable link, whether inline or a block (a card wrapped in a link), and stays clickable on both pages when a page break splits it. External URLs open in the viewer's browser and `#id` links jump to the element. Headings `h1`-`h6` become the PDF's bookmarks, nested by level; set `PdfOptions.Bookmarks = false` to leave them out.

SVG

Inline `<svg>` and SVG files (in `<img>` and as `url()` backgrounds, including `data:image/svg+xml` URIs) with `path` (all commands, arcs included), `rect` (incl. rounded), `circle`, `ellipse`, `line`, `polyline`, `polygon`, `text` and `tspan` (`x`, `y`, `dx`, `dy`, `text-anchor`, `dominant-baseline`, `direction`, registered fonts and right-to-left text), `g`, `use`, nested `svg`, `image` (raster images and SVG files, with `preserveAspectRatio`; inside an SVG file only `data:` URIs load, as in browsers), `viewBox`, `preserveAspectRatio` and `transform`, drawn as vector paths. `linearGradient` and `radialGradient` (both units, `gradientTransform`, `spreadMethod`, `href` inheritance) become PDF shadings, and `pattern` fills and strokes (both units, `viewBox`, `patternTransform`, `href` inheritance)

become PDF tiling patterns. `clipPath` clips the elements that reference it, and `mask` (luminance or alpha, both units) becomes a PDF soft mask. `marker` draws arrowheads and dots at the vertices of paths, lines, polylines and polygons (`orient` incl. `auto` and `auto-start-reverse`, `markerUnits`, `viewBox`, `refX`/`refY`, clipped to the marker unless `overflow="visible"`). `filter` is applied in pixels, at 144 dpi, and drawn as an image, with every filter primitive: `feGaussianBlur`, `feOffset`, `feDropShadow`, `feFlood`, `feMerge`, `feComposite` (all operators), `feBlend` (separable modes), `feColorMatrix`, `feComponentTransfer`, `feMorphology`, `feTurbulence`, `feImage`, `feTile`, `feConvolveMatrix`, `feDisplacementMap`, and `feDiffuseLighting` and `feSpecularLighting` with distant, point and spot lights and `lighting-color`, in linearRGB or sRGB. Primitive subregions (`x`, `y`, `width`, `height`) work as in Chrome. Text under a filter stays selectable and searchable: an invisible copy of it is kept on top of the image. As in browsers, an SVG file is styled only by its own `<style>` elements and attributes, not by the page's CSS, and loads nothing else.

Images

JPEG (embedded as-is, RGB, grayscale and CMYK), PNG (all colour types and bit depths, transparency, interlacing), GIF (transparency, interlacing), WebP (lossy, lossless and with alpha), BMP (1 to 32 bits, palettes, RLE, bit fields and alpha) and TIFF (uncompressed, LZW, Deflate, PackBits, CCITT G3 and G4, JPEG; strips or tiles, planar, alpha, 16-bit; bilevel images stay 1 bit per pixel), from files, `data:` URIs or your `ResourceResolver`. Animated GIF and WebP files show their first frame, as a PDF page is still. SVG files are drawn as vectors (see SVG above). Opaque PNGs are passed through without re-encoding. Images wider than the page are scaled down to fit.

Paged media

> **Changed in 1.9.0:** Page rules and margin boxes, Footnotes.

Page rules and margin boxes Changed in 1.9.0

`@page` rules set the page size, margins and background, and their margin boxes (`@top-left`, `@top-center`, `@bottom-right` and the other thirteen) print headers and footers with `counter(page)` and `counter(pages)`, chosen per page with `:first`, `:left` and `:right`. Those three can also change the page's size, margins and background: `@page :first { margin-top: 70mm }` leaves room for a letterhead on page 1 only, and mirrored `:left` and `:right` margins give a bound booklet a wider inner margin, and each page is laid out at its own content width. The first page is a right page; in a right-to-left document (`dir="rtl"` on `<html>`) it is a left page, and PDF viewers are told the pages run right to left. On top of that:

- **Named pages**: `page: wide` puts an element on pages styled by `@page wide { ... }`, with their own size, orientation, margins, background and margin boxes. Content on another named page starts a new page, so a landscape table or a full-bleed cover can sit between portrait pages.
- **Running headers**: `string-set` copies text from the document (`content()`, `attr()`, strings and counters) into a named string, and `string(name)` shows it in a margin box: the chapter title in the page header, say. As in the spec, `string(name, first | start | last | first-except)` picks which value a page shows.
- **Tables of contents**: `target-counter(attr(href), page)` shows the page number of the element a link points to, and `leader(dotted)` (or `solid`, `space`, or a string) fills the space before it. `target-text()` shows the linked element's text. The document is laid out again with the numbers it finds, so they are always right.

```
@page { size: A4; margin: 20mm;
  @top-center { content: string(chapter) }
  @bottom-right { content: "Page " counter(page) " of " counter(pages) } }
@page cover { margin: 0; background: #1d3557; @top-center { content: none } @bottom-right {
content: none } }
@page wide { size: A4 landscape }
.cover { page: cover }
.wide-table { page: wide }
h2 { string-set: chapter content(); break-before: page }
nav a::after { content: leader(dotted) target-counter(attr(href), page) }
```

Chrome supports named pages and margin boxes, but not running strings, `target-counter()` or `leader()`.

Footnotes New in 1.9.0

`float: footnote` moves an element to the footnote area at the bottom of the page its call is on, as in CSS Generated Content for Paged Media. The `footnote` counter numbers them; `::footnote-call` styles the number in the text (a superscript by default) and `::footnote-marker` the number before the note. `@page { counter-reset: footnote }` restarts the numbering on every page, and `@footnote` inside `@page` styles the area with margins, padding, borders, a background and `max-height`. `footnote-display: inline` runs short notes together on one line, and `footnote-policy: line` moves the line with the call to the next page when its note does not fit. A long note continues on the next page. Each call links to its note, and tagged PDFs mark the notes as `Note` (PDF/UA-1) or `FENote` (PDF/UA-2).

```
.note { float: footnote; font-size: 8pt }
.note::footnote-call { content: "[" counter(footnote) "]"; vertical-align: baseline }
@page { @footnote { border-top: 0.5pt solid #999; padding-top: 4pt; margin-top: 8pt } }
```

```
<p>Paid on the last banking day<span class="note">Or the day before a holiday.</span>.</p>
```

Running header and footer from HTML

An element with the `data-sticky-pdf-header` or `data-sticky-pdf-footer` attribute is taken out of the flow and drawn at the top or bottom of every page, and the page content flows in the space between. Inside it, `{page}` and `{total-pages}` (or `{pages}`) are replaced with the page number and page count. An optional `height` attribute (`height="17mm"`) reserves at least that much space.

```
<header data-sticky-pdf-header height="17mm">ACME Ltd · Payslip</header>
<main>...</main>
<footer data-sticky-pdf-footer height="12mm">Page {page} of {total-pages}</footer>
```

Markdown

Markdown goes to PDF through the same engine as HTML: it is turned into HTML with a GitHub-like stylesheet first, so everything on HTML and CSS support applies to it. You can also stop at the HTML.

```
// Markdown to PDF, styled like a GitHub README
byte[] pdf = MarkdownToPdf.Convert(File.ReadAllText("README.md"));

// File to file; relative image paths resolve against the Markdown file's folder
MarkdownToPdf.ConvertFile("README.md", "README.pdf", new PdfOptions { FooterText = "Page
{page} of {pages}" });

// Markdown to HTML: a body fragment, or a complete page with the stylesheet
string fragment = MarkdownToHtml.Convert("# Hello\n\n*World*");     // <h1
id="hello">Hello</h1>\n<p><em>World</em></p>
string page = MarkdownToHtml.ConvertDocument(markdown);
MarkdownToHtml.ConvertFile("README.md", "README.html");
```

`MarkdownOptions` switches the GitHub extensions off for strict CommonMark (`GitHubExtensions = false`), drops heading ids (`HeadingIds = false`), shows raw HTML as text (`AllowRawHtml = false`), replaces the stylesheet (`StyleSheet`, see `MarkdownToHtml.DefaultStyleSheet`) or sets the title (default: the first heading). `MarkdownToPdf` takes `PdfOptions` and `MarkdownOptions` side by side.

What is supported

CommonMark 0.31 (651 of the 652 spec examples pass; the one miss is an exotic HTML5 entity name) plus the GitHub Flavored Markdown extensions: tables with column alignment, `~~strikethrough~~`, task list items (`- [x]`) and bare `www.`/`https://`/e-mail autolinks. Headings get GitHub-style ids, so `[see below](#heading-text)` links work in the PDF too. The parser runs in linear time on hostile input (deep nesting, unclosed brackets, long delimiter runs). See Limitations for what is not supported.

Fonts

Register TrueType or OpenType fonts and use them by name in CSS. Only the glyphs a document uses are embedded, and text can use any Unicode character the fonts have (Greek, Cyrillic, Central European, CJK, ...):

```
var options = new PdfOptions();
options.Fonts.AddFile("fonts/OpenSans-Regular.ttf")
             .AddFile("fonts/OpenSans-Bold.ttf");            // family, weight and style are
read from the files
options.Fonts.Add(bytes, familyName: "sans-serif");          // or override a generic family
options.Fonts.AddDirectory(@"C:\Windows\Fonts");              // read once, loaded only when
used
byte[] pdf = HtmlToPdf.Convert("<p style=\"font-family: 'Open Sans'\">Łódź, Ελλάδα,
Москва</p>", options);
```

@font-face

Documents can bring their own fonts with `@font-face` (files, `data:` URIs or `ResourceResolver` URLs; `local()` refers to registered fonts):

```
@font-face { font-family: Brand; src: url(fonts/brand.woff2) format("woff2"); font-weight:
400; }
@font-face { font-family: Brand; src: url(fonts/brand-bold.ttf); font-weight: 700; unicode-
range: U+0000-024F; }
```

A `FontCollection` can be shared between options and parallel conversions. Formats: `.ttf`, `.otf`, `.ttc` collections, WOFF and WOFF2 (including WOFF2 collections; decoded with .NET's built-in Brotli). Fonts are matched like browsers do: by family, then style, then the nearest weight; a missing bold or italic face is synthesized. A character that the chosen font lacks is drawn with the next font in the `font-family` list that has it, then the standard font, then any other registered font.

Variable fonts

Variable fonts (one file with a weight, width, slant or other axis, with TrueType or CFF2 outlines) work with the usual CSS properties. PdfPecker builds a static instance of the font, with its kerning, for each combination a document uses and embeds a subset of it, so bold text uses the font's real bold rather than a synthesized one:

```
@font-face { font-family: Inter; src: url(fonts/InterVariable.ttf); }    /* covers all of the
font's weights */
h1 { font-family: Inter; font-weight: 750; }
.narrow { font-stretch: 75%; }                                           /* the wdth axis */
em { font-style: italic; }                                               /* the ital axis, or
slnt */
.custom { font-variation-settings: "wght" 650, "GRAD" 50; }              /* any axis,
overrides the rest */
```

A variable `@font-face` without `font-weight` covers the font's whole weight range, and so does a variable font added through `options.Fonts`. A face declared with a single weight (`font-weight: 400`) stays at that weight, and bold is synthesized as for any other font. The optical size axis (`opsz`) follows the font size unless `font-optical-sizing: none` is set. CFF2 fonts are embedded as ordinary CFF fonts, since few PDF readers can draw CFF2.

OpenType features

`font-variant-numeric` and `font-feature-settings` turn on a font's optional glyphs, as in a browser. Tabular figures line up amounts in a column:

```
td.amount { font-variant-numeric: tabular-nums; }          /* every digit the same width */
.oldstyle { font-variant-numeric: oldstyle-nums slashed-zero; }
.alt { font-feature-settings: "ss01", "salt" 2; }           /* stylistic set 1, the second
alternate */
```

Features that swap one glyph for another (`tnum`, `pnum`, `lnum`, `onum`, `zero`, `ss01` to `ss20`, `salt`, `case` and the like) are applied; the font must have them. Features that depend on the surrounding characters, such as `diagonal-fractions` and `ordinal` in most fonts, are not. Text copied from the PDF keeps the original characters.

Standard fonts

Text without a registered font uses the standard PDF fonts (Helvetica, Times, Courier), which viewers supply themselves, so files stay small but are limited to Western European characters. Set `EmbedStandardFonts = true` to embed the bundled Liberation fonts in their place instead: they have the same widths, so the layout does not change, the text looks the same in every viewer, and any character they cover (Latin, Greek, Cyrillic, ...) works. PDF/A always does this. It adds a subset of each font used to the file (typically 10-25 KB per font).

Missing fonts and characters

A family that no font provides falls back silently, like in a browser. To notice, set `MissingFont`: it is called once per conversion for each family that had to be substituted, and throwing from it fails the conversion. To use the fonts installed on the machine for families you have not registered, set `UseSystemFonts = true` (off by default, so the output does not depend on the machine):

```
options.MissingFont = family => logger.LogWarning("Font {Family} is not available", family);
options.UseSystemFonts = true;   // looks in C:\Windows\Fonts, /Library/Fonts,
/usr/share/fonts, ...
```

Likewise, a character that no font has is drawn as an empty box (blank space in PDF/A). `MissingCharacter` is called once for each such character before anything is written, so a test can make sure a template's fonts cover its data:

```
options.MissingCharacter = ch => throw new InvalidOperationException($"No font has
U+{char.ConvertToUtf32(ch, 0):X4}");
```

How fonts are embedded

Registered and `@font-face` TrueType/OpenType fonts are embedded as subsets (Type0/CIDFontType2 for TrueType outlines, CIDFontType0 with an OpenType program for CFF outlines) with a ToUnicode map, so text can be searched and copied. Without them, the PDF standard fonts are used: `font-family` maps to Helvetica (sans-serif, Arial, Verdana, ...), Times (serif, Georgia, ...) or Courier (monospace, Consolas, ...), each in regular, bold, italic and bold italic. With `EmbedStandardFonts` or PDF/A, these are replaced by embedded subsets of the bundled, metric-compatible Liberation Sans, Serif and Mono.

Right-to-left text, CJK and hyphenation

> **Changed in 1.9.0:** Chinese, Japanese and Korean line breaking.

Right-to-left text: Arabic and Hebrew

Arabic and Hebrew are laid out right to left and shaped like browsers do, with PdfPecker's own implementation of the Unicode Bidirectional Algorithm (UAX #9, passing all of Unicode's conformance tests) and of OpenType shaping (GSUB and GPOS, following HarfBuzz). Arabic letters join, lam-alef and other ligatures form, and harakat and Hebrew points are placed on their letters by the font's mark positioning. Mixed text (an English product name or a number inside an Arabic sentence) is put in the right order, and brackets are mirrored. Register a font that has the script, then set the direction with the `dir` attribute or CSS `direction`:

```
var options = new PdfOptions();
options.Fonts.AddFile("fonts/NotoNaskhArabic-Regular.ttf");
byte[] pdf = HtmlToPdf.Convert("""
    <html dir="rtl" lang="ar"><body style="font-family: 'Noto Naskh Arabic'">
      <h1>فاتورة رقم ٢٠٢٦/١٥</h1>
      <p>المبلغ الإجمالي: 1,250.00 DKK</p>
    </body></html>
    """, options);
```

In a right-to-left block, text aligns right (`text-align: start`), and table columns, flex rows, grid columns and list markers run from the right. `dir="auto"`, `<bdi>` and `<bdo>` work as in browsers, and so do `unicode-bidi`, `text-align: start`/`end` and the logical margin and padding properties (`margin-inline-start`, `padding-inline-end`, ...). `text-overflow: ellipsis` and `line-clamp` cut right-to-left lines at their left end. SVG `<text>` is reordered and shaped too, with `direction` deciding which way a text chunk runs from its `x`. Copied or extracted text gives back the original characters in logical order. A font without OpenType Arabic features is shaped with the Unicode presentation forms instead.

Chinese, Japanese and Korean line breaking New in 1.9.0

Chinese and Japanese are written without spaces, so a line may break between almost any two characters. PdfPecker finds those breaks with the Unicode Line Breaking Algorithm (UAX #14) and the Japanese kinsoku rules, as browsers do: a line never starts with closing punctuation such as `、`, `。` or `）`, or with an iteration mark such as `々`, and never ends with an opening bracket or quotation mark. Nothing is drawn at these breaks, and Latin words, numbers and emoji sequences inside the text stay whole. Text in other scripts breaks at spaces and hyphens, as before. The standard and bundled fonts have no CJK characters, so register a font that has them:

```
var options = new PdfOptions();
options.Fonts.AddFile("fonts/NotoSansJP-Regular.ttf");
byte[] pdf = HtmlToPdf.Convert("""
    <html lang="ja"><body style="font-family: 'Noto Sans JP'; line-break: strict">
      <p>                                                    </p>
    </body></html>
    """, options);
```

CSS `line-break` sets how strict the rules are, as in Chrome:

- `auto` (the default) and `strict` keep small kana (, , ...) and the prolonged sound mark off the start of a line.
- `normal` lets small kana and start a line, and allows a break before the hyphens `-`, `–`, and .
- `loose` also allows a break before an iteration mark or centred punctuation such as and , and on both sides of an ellipsis.
- `anywhere` breaks between any two characters and ignores the punctuation rules, in every script.

Korean text breaks between its syllables by default, as in Chrome. `word-break: keep-all` breaks CJK text only at spaces and punctuation instead, which suits Korean, written with spaces between words. `word-break: break-all` and `white-space: nowrap` work as for any other text. A justified line is stretched only at its spaces, so a Chinese or Japanese line without spaces is not justified (see Limitations).

Hyphenation

`hyphens: auto` hyphenates words with the TeX hyphenation patterns (Liang's algorithm) of the text's language: the nearest `lang` attribute, else `PdfOptions.Language`. As in Chrome, a word is hyphenated only where it does not fit on its line, and a hyphen (or the `hyphenate-character`) is shown at the break. Words shorter than five letters, words with digits or other non-letters inside, URLs and e-mail addresses, and capitalized English words are not hyphenated, and a word with `` breaks only there. Patterns for English (US patterns for every `en` tag), Danish, German, Dutch and Norwegian (`nb`, `nn`, `no`) are built in and loaded on first use; others can be supplied as TeX pattern files, such as those of the hyph-utf8 project:

```
var options = new PdfOptions
{
    // Your own patterns for a language (they take precedence over the built-in ones).
    Hyphenation = new Dictionary<string, HyphenationPatterns>
    {
        ["sv"] = new HyphenationPatterns(File.ReadAllText("hyph-sv.tex"), leftMin: 2,
rightMin: 2),
    },
};
byte[] pdf = HtmlToPdf.Convert("""
    <html lang="da"><body style="hyphens: auto; text-align: justify">
      <p>Arbejdsmarkedsbidraget betales af lønmodtageren.</p>
      <p lang="sv">Avstavning med egna mönster.</p>
    </body></html>
    """, options);
```

PDF/A, passwords and watermarks

Watermarks and passwords

```
var options = new PdfOptions
{
    // A diagonal stamp on every page. Also: Image (PNG, JPEG, GIF, WebP, BMP, TIFF or SVG
bytes; SVG stays vector), FontSize, LetterSpacing, Angle, Opacity, Bold, Behind.
    Watermark = new PdfWatermark { Text = "WATERMARK", Color = "#c00000", Opacity = 0.25f },

    // AES-256 encryption (PDF 2.0 / Acrobat X and later). Without an OwnerPassword a random
one is used,
    // so the permissions cannot be lifted.
    Security = new PdfSecurity
    {
        UserPassword = "1234",                                  // needed to open the file;
null = opens freely
        OwnerPassword = "owner-secret",                         // full access
        Permissions = PdfPermissions.Print | PdfPermissions.Copy,
    },
};
```

PDF/A

For archiving, set `Conformance = PdfConformance.PdfA2b` (or another level below): the PDF gets XMP metadata, an sRGB output intent and a file ID, and CMYK JPEGs are converted to RGB. All fonts are embedded: text without a registered font uses the bundled Liberation fonts (see Fonts). PDF/A does not allow encryption, so combining `Conformance` with `Security` throws an `ArgumentException`.

```
var options = new PdfOptions { Conformance = PdfConformance.PdfA4 };
```

Level	Standard	File	Use it for
`PdfA2b`, `PdfA3b`	ISO 19005-2/-3, level B	PDF 1.7	the usual archive format; part 3 also allows attachments (e-invoices)
`PdfA2u`, `PdfA3u`	ISO 19005-2/-3, level U	PDF 1.7	as level B, plus a Unicode mapping for every character, so text can be searched and copied
`PdfA4`	ISO 19005-4:2020	PDF 2.0	the current part; identified in the XMP only (`pdfaid:rev` 2020, no conformance letter), no document information dictionary
`PdfA4f`	ISO 19005-4:2020, with files	PDF 2.0	PDF/A-4 with attachments: only with `EInvoice` (it throws without one)

Every font is embedded with a ToUnicode map, and a character no font has is left out as blank space rather than drawn as `.notdef` (in SVG text too), so levels U cost nothing over B. A PDF 2.0 file keeps the title, author, keywords and custom properties in its XMP metadata, since PDF 2.0 deprecates the information dictionary. Every level, on its own and with PDF/UA, passes veraPDF 1.28 (flavours 2b, 2u, 3b, 3u, 4 and 4f) on documents with text, tables, lists, images, links, form fields, SVG and Markdown.

AES-256 keys come from a password hash that is slow on purpose, which adds about 3 ms to a conversion.

Passwords on existing PDFs

To add, change or remove the password of a PDF you already have (from any producer, not only PdfPecker), use `PdfProtector`:

```
// Encrypt any PDF with AES-256
byte[] locked = PdfProtector.Encrypt(pdf, new PdfSecurity { UserPassword = "1234", Permissions
= PdfPermissions.Print });

// Change the passwords or permissions of an encrypted PDF (give its current user or owner
password)
byte[] relocked = PdfProtector.Encrypt(locked, new PdfSecurity { UserPassword = "5678" },
password: "1234");

// Remove the encryption, and with it the permission restrictions
byte[] open = PdfProtector.Decrypt(locked, "1234");

// Stream overloads, and a check
PdfProtector.Encrypt(inputStream, outputStream, security);
bool encrypted = PdfProtector.IsEncrypted(pdf);
```

Encrypted inputs can use RC4 (40 to 128 bit), AES-128 or AES-256; the output of `Encrypt` is always AES-256. Like other PDF tools, `PdfProtector` accepts the user password as well as the owner password (or none, for a file that only restricts permissions), so anyone who can open a document can also lift its permissions. A wrong password throws `PdfPasswordException`. The document is rewritten as a whole and keeps its pages, bookmarks, links, form fields and values, tags, metadata, page labels and attachments; earlier revisions from incremental updates are dropped. Signed PDFs are refused with an `ArgumentException` (rewriting them would break the signatures; sign after encrypting instead), as is encrypting a PDF/A file, or a PDF/UA file without `PdfPermissions.ExtractForAccessibility`. `Decrypt` returns a PDF that is not encrypted unchanged.

`Watermark` applies while converting HTML; to stamp a PDF you already have, see Stamping PDFs.

Accessibility: tagged PDF and PDF/UA

```
var options = new PdfOptions
{
    Accessibility = PdfAccessibility.PdfUA1,   // or PdfUA2, or Tagged: the same structure
without the PDF/UA claim
    Language = "da",                           // default: <html lang="..."> ; for PDF/UA
without either, "en"
    Conformance = PdfConformance.PdfA2b,       // optional: PDF/UA-1 combines with PDF/A-2 and
3, PDF/UA-2 with PDF/A-4
};
```

Tagged PDF

`Tagged` gives the PDF a structure tree built from the HTML, so screen readers, reflow and text extraction follow the document rather than the page layout: headings (`H1` to `H6`), paragraphs, lists with their labels, tables with header scopes and spans, figures with their alt text, links (tied to their clickable areas), block quotes, code and sections, all in document order, with `lang` attributes as the language of the element. Text outside a paragraph (straight in a `div`, say) gets one. Backgrounds, borders, running headers and footers, `HeaderText` / `FooterText`, watermarks, repeated table headers, fixed boxes and decorative images are artifacts that assistive technology skips.

PDF/UA-1

`PdfUA1` adds what ISO 14289-1 asks beyond the tags: every font embedded (the bundled Liberation fonts stand in for the standard ones), the title shown in the viewer's title bar and the PDF/UA identification in the XMP metadata. The output passes veraPDF's PDF/UA-1 checks, on its own and together with PDF/A-2b or PDF/A-3b (e-invoices included). It needs a title: `Title`, the `<title>` element or else the first heading; without one, conversion throws `ArgumentException`, as it does when `Security` leaves out `PdfPermissions.ExtractForAccessibility`.

PDF/UA-2

`PdfUA2` (ISO 14289-2:2024) is the PDF 2.0 successor: the file is written as PDF 2.0, its tags are in the PDF 2.0 structure namespace (block quotes, quotes and code, which PDF 2.0 dropped, keep their PDF 1.7 type and are role mapped to `Div` and `Span`), links and bookmarks go to structure destinations (the element they point at, not only its page), lists state their numbering and form fields describe themselves, and the XMP says `pdfuaid:part` 2 with `pdfuaid:rev` 2024. It has the same title and language requirements as `PdfUA1` and passes veraPDF's PDF/UA-2 checks, on its own and with `PdfA4`. PDF/UA-1 and PDF/A-2 and 3 are based on PDF 1.7, PDF/UA-2 and PDF/A-4 on PDF 2.0, so `PdfUA2` with `PdfA2b` (or any PDF/A-2 or 3 level) and `PdfUA1` with `PdfA4` throw `ArgumentException`.

What your HTML needs

What the tags can't fix is up to the HTML: give images an `alt` (an image without `alt`, `aria-label` or `title`, or an `<svg>` without a `<title>`, counts as decorative), use real headings, lists and `th` cells, and mark purely visual content `aria-hidden="true"`. Heading levels that skip one (`h1` then `h3`) are tagged one level below the previous heading, as PDF/UA does not allow gaps.

Fillable forms

> **Changed in 1.9.0:** Filling and flattening existing PDFs.

Form controls become fillable PDF form fields (AcroForm), drawn as Chrome prints them and filled with their current values. Adobe Reader, Preview, Chrome, Firefox and other viewers let the reader fill them in and save the result. The fields of an existing PDF can be read, filled and flattened too.

```
var html = """
    <p>Name <input name="name" value="Jane Doe" required></p>
    <p><label><input type="checkbox" name="newsletter" checked> Send me the
newsletter</label></p>
    <p><input type="radio" name="size" value="S"> S <input type="radio" name="size" value="M"
checked> M</p>
    <p><select name="country"><option value="dk">Denmark</option><option
value="se">Sweden</option></select></p>
    <p><textarea name="comments" rows="4" cols="40"></textarea></p>
    """;
byte[] pdf = HtmlToPdf.Convert(html);

// Or print the controls as they are, without form fields:
byte[] flat = HtmlToPdf.Convert(html, new PdfOptions { FormFields = false });
```

HTML	PDF field
`<input>` of type `text`, `email`, `tel`, `url`, `search`, `number`, `date` and the like	Text field (`maxlength` sets the maximum length)
`<input type="password">`	Password text field; the value is drawn as dots and not stored
`<textarea>`	Multi-line text field
`<input type="checkbox">`	Check box; its `value` is the export value (default `Yes`)
`<input type="radio">`	Radio buttons: those with the same `name` are one field, and each `value` is a choice
`<select>`	Drop-down list (combo box); with `multiple` or `size` > 1 a list box, multi-select with `multiple`
`<button>`, `<input type="submit">`, `reset`, `button`	Drawn only: a PDF has nothing to submit

Field names and behaviour

Fields are named after the `name` attribute (a period becomes `_`, since PDF uses it to nest fields; a repeated name gets `_2`, `_3`, ...; a control without a name gets `text1`, `checkbox1`, ...). `readonly` and `required` set the field's flags, `title`, `aria-label` or its `<label>` becomes its tooltip, and a `placeholder` is drawn in grey while the field is empty. Disabled controls (or controls in a disabled `<fieldset>`) are drawn but not fillable, and hidden inputs are left out. Each field has its own appearance, so the PDF looks right before anyone touches it and in viewers that never redraw fields; while a field is edited, the viewer draws its text in Helvetica, Courier or Times at the control's size and colour. Forms work with `Security` (the values are encrypted with the rest), `Conformance` (veraPDF

passes the fields as PDF/A-2b and 3b), `Accessibility` (each widget sits in a `Form` element described by its label, title or name, and veraPDF passes PDF/UA-1), `Signature` (the signature field joins the form) and `PdfMerger` (repeated names are renamed).

Filling and flattening existing PDFs New in 1.9.0

`PdfForm` reads, fills and flattens the form fields of any PDF, not only those PdfPecker made, such as a government form or a contract template from another tool:

```
// The fields, by fully qualified name ("address.city" for a nested one).
foreach (PdfFormField field in PdfForm.GetFields(pdf))
    Console.WriteLine($"{field.Name} ({field.Type}) = {field.Value}, choices: {string.Join(",
", field.Options)}");

var values = new Dictionary<string, string>
{
    ["name"] = "Jane Doe",
    ["address.city"] = "Copenhagen",
    ["newsletter"] = "true",       // a check box: its on state, or true/yes/on/1;
Off/false/no/0/"" clears it
    ["size"] = "M",                // a radio button group: the state name of one button
    ["country"] = "dk",            // a combo box or list box: a choice's value or its text
};
byte[] filled = PdfForm.Fill(pdf, values);

// Fill and flatten in one go, so the values can no longer be changed; or flatten a PDF as it
is.
byte[] final = PdfForm.Fill(pdf, values, new PdfFormFillOptions { Flatten = true });
byte[] flat = PdfForm.Flatten(filled);
PdfForm.FillFile("application.pdf", "application-filled.pdf", values);
```

`PdfFormField` gives each field's `Type` (`Text`, `CheckBox`, `RadioButton`, `ComboBox`, `ListBox`, `PushButton` or `Signature`), its `Value` (and `Values` for a list box with several choices), the `Options` it accepts, its flags (`ReadOnly`, `Required`, `Multiline`, `MaxLength`, ...), its tooltip and the page it is on. A name that is no field, a choice the field does not have, or text longer than `MaxLength` throws `ArgumentException`; set `IgnoreUnknownFields` to skip names the PDF does not have. Push buttons and signature fields cannot be filled (sign the latter with `PdfSigner`).

Every filled field gets a new appearance in the field's own font, size, colour and alignment, with automatic font size, multi-line text and comb fields laid out, and the box and background of its old appearance kept, so the values show in every viewer and in `PdfToImage`. When the field's font cannot show a value, a subset of `PdfFormFillOptions.Font` (by default the bundled Liberation Sans, which covers Latin, Greek and Cyrillic) is embedded for it. XFA form data, which would override the filled fields in some viewers, is removed.

A signed or encrypted PDF is filled with an incremental update: the original bytes stay as they were, so the signatures stay valid (when their permissions allow form filling) and the encryption is kept; pass the password in `PdfFormFillOptions.Password`. Other PDFs are written out in full, without the old values in earlier revisions. Flattening draws each field into its page and removes it, keeping links, comments and other annotations; it changes the pages, so a signed PDF is refused with `InvalidOperationException`.

Digital signatures

> **Changed in 1.9.0:** External signers: HSMs, smart cards and key vaults, Verifying signatures, Long-term validation.

Sign the PDF with a certificate (a `.pfx` file or one from the certificate store). Adobe Reader then shows who signed it and whether the file was changed afterwards. The key can also stay in an HSM, smart card or cloud key vault (external signers), and the signatures of any PDF can be verified and made verifiable for years.

```
var cert = X509CertificateLoader.LoadPkcs12FromFile("signer.pfx", "pfx-password");
var options = new PdfOptions
{
    Signature = new PdfSignatureOptions(cert)
    {
        Reason = "Approved",
        Location = "Copenhagen",
        // Optional: a visible box (points from the top-left corner; Page = -1 is the last
page).
        Appearance = new PdfSignatureAppearance { Page = -1, X = 340, Y = 720, Width = 220,
Height = 70 },
        // Optional: a trusted RFC 3161 timestamp (PAdES B-T). The only option that uses the
network.
        TimestampUrl = "http://timestamp.digicert.com",
    },
};

// Or sign any existing PDF; earlier signatures stay valid.
byte[] signed = PdfSigner.Sign(File.ReadAllBytes("in.pdf"), new PdfSignatureOptions(cert));

// A contract template with empty signature fields: sign into one by name, at its place on the
page.
byte[] countersigned = PdfSigner.Sign(contract, new PdfSignatureOptions(cert) { FieldName =
"Buyer" });
```

Signing into an existing field

When `FieldName` names an empty signature field the PDF already has (`"Parties.Buyer"` for a nested one), the signature goes into it and the box fills the field's rectangle; signing a field of another type or one that is already signed throws `PdfSignatureException`. Any other name adds a new field.

Formats and compatibility

Signatures are PAdES baseline (`ETSI.CAdES.detached`) by default, or classic `adbe.pkcs7.detached` with `Format = PdfSignatureFormat.Pkcs7Detached`; RSA and ECDSA keys with SHA-256/384/512. The signature is appended as an incremental update, and it works together with `Security` (encrypted files) and `Conformance` (PDF/A: the visible box is drawn as outlines, so it needs no unembedded font). The CMS structure is built with .NET's own `System.Formats.Asn1` and

`System.Security.Cryptography`; no extra package is needed. Signing adds a few milliseconds (the RSA or ECDSA operation and re-reading the file); conversions without a signature are unaffected.

External signers: HSMs, smart cards and key vaults New in 1.9.0

When the private key cannot leave its hardware security module, smart card, PKCS#11 token or cloud key vault, pass a `PdfExternalSigner` with the certificate. PdfPecker builds the PDF and the CMS structure, and asks your callback only for the raw signature value. The certificate needs only its public part. The callback gets a `PdfSignatureRequest` with `Digest` for "sign hash" APIs (Azure Key Vault `SignAsync`, AWS KMS `Sign` with `MessageType.DIGEST`, `RSA.SignHash`) and `Data` for APIs that hash themselves; both give the same signature.

```
// The certificate without its private key; the key stays in Azure Key Vault.
var cert = X509CertificateLoader.LoadCertificateFromFile("signer.cer");
var keyVault = new CryptographyClient(keyId, new DefaultAzureCredential());

var signer = new PdfExternalSigner(PdfSignatureAlgorithm.RsaPkcs1, async (request, ct) =>
    (await keyVault.SignAsync(SignatureAlgorithm.RS256, request.Digest.ToArray(),
ct)).Signature);
byte[] signed = await PdfSigner.SignAsync(pdf, new PdfSignatureOptions(cert, signer) { Reason
= "Approved" });

// A synchronous callback, here with a key .NET can use directly (such as one from a PKCS#11
wrapper).
var local = new PdfExternalSigner(PdfSignatureAlgorithm.RsaPss,
    request => rsa.SignHash(request.Digest.ToArray(), request.HashAlgorithm,
RSASignaturePadding.Pss));
byte[] signed2 = PdfSigner.Sign(pdf, new PdfSignatureOptions(cert, local));
```

`PdfSignatureAlgorithm` is `RsaPkcs1` (RS256 and the like), `RsaPss` (PS256, with a salt as long as the hash) or `Ecdsa` (ES256, DER-encoded or raw r||s); the hash is `PdfSignatureOptions.HashAlgorithm`, SHA-256 by default. The returned signature is checked against the certificate before it goes into the PDF, so a wrong key or padding throws `PdfSignatureException` instead of producing a broken file. The callback is normally called once per signature (twice in the rare case the signature does not fit the space reserved for it). `PdfSigner.SignAsync` awaits an asynchronous callback and the timestamp server instead of blocking a thread; `PdfSigner.Sign` and signing during `HtmlToPdf.Convert` with `PdfOptions.Signature` wait for it. Everything else, visible boxes, timestamps and existing fields, works as with a certificate's own key.

Verifying signatures New in 1.9.0

`PdfSignatureVerifier` checks the signatures and document timestamps of any PDF, oldest first, and reports what it finds rather than throwing for a broken signature:

```
foreach (PdfSignatureVerification s in PdfSignatureVerifier.Verify(pdf))
{
    Console.WriteLine($"{s.FieldName}: signed by {s.SignerName} at {s.TimestampTime ??
s.SigningTime}");
    Console.WriteLine($"  valid: {s.Valid}, intact: {s.IsIntact}, covers the whole file:
{s.CoversWholeDocument}");
    foreach (string problem in s.Problems) Console.WriteLine("  " + problem);
}

// Trust only your own CA instead of the operating system's roots, and check revocation
online.
var results = PdfSignatureVerifier.VerifyFile("contract.pdf", new
PdfSignatureVerificationOptions
{
    TrustedRoots = new X509Certificate2Collection(companyRootCert),
    RevocationMode = X509RevocationMode.Online,
});
```

`IsIntact` means the signed bytes are unchanged (`DigestMatches`), the byte range is well formed (`ByteRangeValid`), the signature value verifies with the signer's certificate (`SignatureValid`) and an embedded timestamp, if any, is sound (`TimestampValid`). `Valid` adds that the certificate chains to a trusted root at the time of the timestamp, else the claimed signing time (`ChainTrusted`, with `ChainErrors` saying why not). Chains are built from the certificates in the signature, the PDF's Document Security Store and `ExtraCertificates`. Verification is offline by default: `RevocationMode` is `NoCheck` unless you set it, and `VerificationTime` checks the chain at another time.

A signature followed by later changes stays valid, because that is how a second signature or validation data is added: `UpdatesAfterSigning` counts the incremental updates after it, and `CoversWholeDocument` is false. What those updates change is not analysed (see Limitations), so check `CoversWholeDocument` on the last signature when nothing may follow it. PAdES, `adbe.pkcs7.detached`, `adbe.pkcs7.sha1` and document timestamps (`ETSI.RFC3161`, `IsDocumentTimestamp`) are read, from PDFs made by any tool.

Long-term validation New in 1.9.0

A signature can normally be checked only while its certificates are valid and their CA answers. `PdfSigner.AddValidationData` makes it checkable for years: it adds a Document Security Store with every certificate in the chains of the PDF's signatures and timestamps, and the OCSP responses and CRLs that show none was revoked (PAdES B-LT). With `TimestampUrl` a document timestamp follows, which seals the signatures and the validation data with a trusted time (PAdES B-LTA). Both are incremental updates, so the signatures stay valid.

```
byte[] signed = PdfSigner.Sign(pdf, new PdfSignatureOptions(cert) { TimestampUrl =
"http://timestamp.digicert.com" });

byte[] ltv = PdfSigner.AddValidationData(signed, new PdfLtvOptions
{
    FetchRevocationData = true,                      // ask each certificate's OCSP
responder, else its CRL
    TimestampUrl = "http://timestamp.digicert.com",   // PAdES B-LTA
});

// Offline: only the chains, completed with your own certificates and revocation data.
byte[] offline = PdfSigner.AddValidationData(signed, new PdfLtvOptions
{
    Certificates = [intermediateCert, rootCert],
    OcspResponses = [ocspResponseDer],
});
```

Without `FetchRevocationData` or `TimestampUrl` nothing goes over the network; OCSP and CRL requests use only the http and https addresses in the certificates, through `HttpClient` if you set one. To make the document timestamp itself long-term verifiable, call `AddValidationData` again later without a `TimestampUrl`. A PDF without signatures throws `PdfSignatureException`.

E-invoices

Factur-X and ZUGFeRD

Pass the invoice's CII XML (from your invoicing system) and PdfPecker attaches it to the PDF, making a Factur-X / ZUGFeRD e-invoice: the pages are the human-readable invoice and the XML is what accounting software reads.

```
var options = new PdfOptions
{
    EInvoice = new PdfEInvoice(File.ReadAllBytes("invoice.xml")),
};
File.WriteAllBytes("invoice.pdf", HtmlToPdf.Convert(invoiceHtml, options));
```

The file is PDF/A-3b, so `Conformance` is raised to `PdfA3b` when it is left at `None`; `PdfA3u` and `PdfA4f` (PDF/A-4 with attachments) work too, while the other levels and `Security` throw. Factur-X and ZUGFeRD specify PDF/A-3, so keep the default unless the receiver asks for PDF/A-4f. The XML is embedded unchanged as `factur-x.xml` (`xrechnung.xml` for XRechnung), listed as an associated file, and the XMP metadata gets the Factur-X extension schema with the document type, version and profile. The profile (MINIMUM, BASIC WL, BASIC, EN 16931, EXTENDED or XRECHNUNG) is read from the XML's guideline ID; pass it as the second constructor argument to set it yourself. The relationship to the pages defaults to `Data` for MINIMUM and BASIC WL and `Alternative` otherwise; set `Relationship` to change it. PdfPecker checks that the XML is a well-formed Cross Industry Invoice (UBL is not allowed in Factur-X) but does not validate its content or build it from invoice data: make sure the visible pages show the same invoice. The output passes the Mustang validator (PDF/A-3b via veraPDF, and the XML's schematron). Other files can be attached next to the invoice with `Attachments` (see Attachments).

PDF attachments in UBL e-invoices (Peppol, OIOUBL)

Peppol BIS Billing 3.0, OIOUBL, EHF, XRechnung UBL and other UBL e-invoices are XML only: the XML is the invoice, and the PDF travels inside it as an attachment. `UblInvoice.AttachPdf` adds one to the invoice or credit note your invoicing system made:

```
byte[] pdf = HtmlToPdf.Convert(invoiceHtml);
byte[] xml = UblInvoice.AttachPdf(File.ReadAllBytes("invoice.xml"), pdf, "INV-1001.pdf");
```

The PDF goes in as an `AdditionalDocumentReference` with `EmbeddedDocumentBinaryObject` (`mimeCode="application/pdf"` and the file name), after any attachments already there and where the UBL schema expects it. The rest of the XML, including its indentation, encoding and declaration, is kept as it was. The reference ID defaults to the file name; pass `id` and `description` to set them. For OIOUBL 2.1 (UBL 2.0) the description is written as the `DocumentType` OIOUBL requires, `PDF` by default. The output passes the UBL schemas, the Peppol BIS 3.0 and EN 16931 schematrons, and the OIOUBL 2.1 schematron. PdfPecker does not build or validate the rest of the invoice.

Attachments

Any file can travel inside the PDF, listed in the viewer's attachments panel: the source spreadsheet of a report, the data behind a chart, a signed contract.

```
var options = new PdfOptions { Conformance = PdfConformance.PdfA3b }; // optional
options.Attachments.Add(new PdfAttachment("figures.csv", csvBytes)
{
    Description = "The figures behind the charts",          // default: the file name
    Relationship = PdfAttachmentRelationship.Data,           // Source, Data, Alternative,
Supplement or Unspecified
    // MimeType = "text/csv",                                // default: guessed from the
extension
    // CreationDate = ..., ModDate = ...                     // default: ModDate is the time
of conversion
});
options.Attachments.Add(PdfAttachment.FromFile("contract.pdf"));
byte[] pdf = HtmlToPdf.Convert(html, options);

// Read them back, or attach files to a PDF made elsewhere.
foreach (PdfAttachment a in PdfAttachments.List(pdf)) Console.WriteLine($"{a.FileName}:
{a.Data.Length} bytes");
byte[]? csv = PdfAttachments.Extract(pdf, "figures.csv");
byte[] more = PdfAttachments.Add(File.ReadAllBytes("scan.pdf"), new PdfAttachment("notes.txt",
notes));
```

How files are stored

Each file is written with its MIME type, size, dates and MD5 checksum, in the document's embedded-files list sorted by name, and as an associated file (`/AF`) with its relationship to the pages, as PDF/A-3 requires. File names must be unique (`ArgumentException` otherwise; an attachment can't take the e-invoice's `factur-x.xml`). Attachments don't change `Conformance`: without one the file is a plain PDF, PDF/A-3b, PDF/A-3u and PDF/A-4f allow any attachment, and PDF/A-2b, PDF/A-2u and PDF/A-4, which allow only PDF/A files as attachments, throw `ArgumentException` rather than produce a file that fails validation. With `Security` the attachments are encrypted with the rest of the file, and for PDF/UA each one gets the description ISO 14289 asks for. The output passes veraPDF's PDF/A-3b, PDF/A-3u, PDF/A-4f, PDF/UA-1 and PDF/UA-2 checks. `PdfAttachments.Add` likewise refuses PDF/A-1, PDF/A-2 and PDF/A-4 files, and adds to PDF/A-3 and PDF/A-4f files.

Adding to an existing PDF

`PdfAttachments.Add` appends the files as an incremental update, so the original bytes, and any digital signatures over them, stay as they were; an encrypted PDF (pass its password) stays encrypted, a PDF/A-3 file gets the new associated files, and PDF/A-1 and PDF/A-2 files are refused. `List` and `Extract` read the embedded-files list and associated files of any PDF, decrypting with the password you pass; files attached to page annotations (the paper-clip icons) are not read.

Merging and splitting PDFs

Merging

`PdfMerger` combines PDFs into one, any PDFs, not only those PdfPecker made:

```
byte[] merged = PdfMerger.Merge(invoicePdf, termsPdf, attachmentPdf);

// With passwords for encrypted inputs (by position), and properties for the result.
byte[] merged2 = PdfMerger.Merge(new[] { a, b }, new PdfMergeOptions
{
    Passwords = new string?[] { null, "secret" },
    Title = "Payslips October",
    CustomProperties = { ["RunId"] = "42" },
});

PdfMerger.MergeFiles("all.pdf", new[] { "a.pdf", "b.pdf" });

// Streams in and out: inputs are read one at a time and the result is written as it is made.
PdfMerger.Merge(new[] { invoiceStream, termsStream }, outputStream);
await PdfMerger.MergeAsync(uploads.Select(f => f.OpenReadStream()), response.Body);
```

Tagged inputs stay tagged: their structure trees are joined under one root, so the merged file is still accessible, and when every input is PDF/UA-1 the result says so too (veraPDF passes it). The merged file is PDF 1.7, so PDF/UA-2 inputs stay tagged but the result claims no PDF/UA level. An input in another language than the first keeps its language on its own part of the tree. Untagged inputs merge as before.

With streams (and `MergeFiles`), only one input is held in memory at a time and the merged document goes to the output as each input is done, so memory follows the largest input rather than the whole batch: merging 100 files of 32 MB in total peaks at 14 MB of managed memory, against 105 MB with byte arrays. `MergeAsync` reads and writes asynchronously, as ASP.NET Core requires for request and response bodies. Input streams are read from their current position and are not disposed; the sequence is enumerated lazily, so an iterator can open each one only when its turn comes. `MergeFiles` writes to a temporary file and moves it into place, so the output can be one of the inputs and a failed merge leaves no partial file behind.

Merging PDFs as they arrive

For PDFs that arrive one at a time from elsewhere, such as downloads read a few at a time, use `StreamingPdfMerger`. It writes each PDF's pages when it is added and then lets the PDF go, so memory stays at about one input, and between inputs you decide what comes next: skip a missing document, or check that anything was added before answering. The output can be an HTTP response body or a file; it does not need to be seekable.

```
var merger = new StreamingPdfMerger(response.Body, new PdfMergeOptions { Title = "Payslips
October" });
foreach (var path in paths)
{
    var pdf = await storage.DownloadAsync(path, ct);
    if (pdf.Length == 0) continue;          // skip a missing document
    await merger.AddAsync(pdf, ct);
}
if (merger.InputCount == 0) return Results.NotFound();
await merger.CompleteAsync(ct);             // writes the page tree, bookmarks, form and
cross-reference table

// Or, from an IAsyncEnumerable<byte[]>:
int pages = await PdfMerger.MergeAsync(DownloadInOrder(paths), response.Body,
cancellationToken: ct);
```

The result is the same as `PdfMerger.Merge`'s, but the output is only a valid PDF once `Complete` has run. If adding a PDF throws (a damaged or encrypted input, `MaxPages` reached), what was written cannot be used and the merger refuses further calls. `Passwords` are matched to inputs in the order they are added.

The pages keep their content, fonts, images, links, annotations, layers and fillable form fields, and each input's bookmarks follow each other. A form field named like one in an earlier input is renamed `name_2`, `name_3` and so on, so the copies keep their own values. Document properties come from the first input unless `PdfMergeOptions` sets them. The result is never encrypted (pass it to `PdfProtector.Encrypt` to protect it) and has no signatures (merging would break them); PDF/A conformance is not carried over. Page labels are kept (pages of an input without them are numbered as in that input), and attachments too, with one named like an earlier input's listed as `name (2).ext`. Images, fonts and other resources that are byte-for-byte the same in several inputs, such as a logo on every payslip of a run, are written once.

Inputs are handled as untrusted: damaged files are repaired where possible, decoding is limited per input, and `MaxPages` (default 100,000) caps the result. Inputs without any pages throw `InvalidDataException`. Merging was tested on PDFs from Chrome (as IronPDF makes them), Ghostscript, LibreOffice, MuPDF, qpdf, ReportLab, pypdf incremental updates, scanners (JPEG, CCITT G4, CMYK, OCR text layers) and damaged files (wrong or missing cross-reference tables, truncated files, junk before the header); every merged page rendered pixel-identical to its original in Poppler. Merging four small documents takes about 1 ms, and 1,000 one-page Chrome payslips about 1 s.

Extracting, reordering and splitting pages

`PdfPages` works on the pages of any existing PDF. Pages are numbered from 1, and ranges are written like `"1-3,5,8-"` (pages 1 to 3, 5, and 8 to the end): `"3,1,2"` reorders, `"1,1"` repeats, `"5-3"` runs backwards and `"last"` is the last page. `PdfPageRange.Parse` reads a range once for reuse; a page past the end throws `ArgumentOutOfRangeException` ("Page 9 is out of range: the PDF has 5 pages.").

```
byte[] summary = PdfPages.Extract(report, "1-3,5");              // or new[] { 3, 1, 2 }
byte[] withoutCover = PdfPages.Remove(report, "1");
byte[] turned = PdfPages.Rotate(scan, 90, "2,4");                 // clockwise; null pages =
every page
byte[][] parts = PdfPages.Split(report, 10);                      // 10 pages each
byte[][] chapters = PdfPages.Split(report, "1-4", "5-9", "10-");
int pages = PdfPages.Count(report);

// Encrypted input, files, streams.
byte[] page = PdfPages.Extract(secret, "2", new PdfPagesOptions { Password = "pw", Title =
"Page 2" });
PdfPages.ExtractFile("report.pdf", "summary.pdf", "1-3");
PdfPages.SplitFile("report.pdf", 1, n => $"report-{n}.pdf");

// Selected pages of several PDFs merged into one (also StreamingPdfMerger.Add(pdf, pages)).
byte[] merged = PdfMerger.Merge(new[] { new PdfMergeInput(invoice, "1"), new
PdfMergeInput(terms, "2-") });
```

For the pages taken, the result keeps their content, links and other annotations, the form fields with a widget on them, the bookmarks that go to them, page labels, the document properties and language. Links and bookmarks to pages left out are dropped (a bookmark with a kept bookmark below it stays, without a destination); links between pages taken go to their new place. A tagged input stays tagged: structure elements are kept when some of their content is on a page taken, so a PDF/UA-1 input gives a PDF/UA-1 result (veraPDF passes extracted, removed and split parts). A page taken more than once is copied with its links, but its form fields stay on the first copy and the later copies are not tagged, so such a result no longer claims PDF/UA. `Split` parses the PDF once for all parts. As with merging, the result is not encrypted and has no signatures.

Stamping PDFs

`PdfStamper` puts a page of one PDF on top of the pages of another, such as a "PAID" or "COPY" stamp, or underneath them, such as company letterhead. Any PDF can be stamped, not only those PdfPecker made:

```
// A one-page stamp PDF, fitted and centred on every page, half transparent.
byte[] paid = PdfStamper.Stamp(invoicePdf, paidStampPdf, new PdfStampOptions { Opacity = 0.5f
});

// Letterhead under the content: the stamp's first page on page 1, its second page on all the
others.
byte[] letter = PdfStamper.Stamp(letterPdf, letterheadPdf, new PdfStampOptions { Underlay =
true, MatchPages = true });

// Actual size in the top right corner, 20 pt in from each edge, on the first page only.
byte[] copy = PdfStamper.Stamp(pdf, copyStampPdf, new PdfStampOptions
{
    Position = PdfStampPosition.TopRight, Scale = 1, OffsetX = -20, OffsetY = 20, Pages = i =>
i == 0,
});

// The stamp as HTML, rendered at each page's size with no margins.
byte[] stamped = PdfStamper.StampHtml(pdf,
    "<div style='position:absolute; right:30pt; top:30pt; border:3pt solid #c00; color:#c00;
font:bold 28pt sans-serif'>PAID</div>");

PdfStamper.StampFile("in.pdf", "stamp.pdf", "out.pdf");
PdfStamper.Stamp(inputStream, stampStream, outputStream);
```

Placement

`StampPage` picks the stamp PDF's page (0-based), `Pages` the pages to stamp (by 0-based index), `Scale` the size (0, the default, fits the stamp page on the page keeping its aspect ratio; 1 is its own size), and `Position`, `OffsetX` (right) and `OffsetY` (down) where it goes. The stamp is placed as the page is displayed, so it is upright on rotated pages and sits in the crop box. The stamp page is copied in once as a form XObject, so its text stays text and its graphics stay sharp, however many pages show it. The page's own content is wrapped so a graphics state it leaves behind cannot shift or recolour an overlay. Underlays are hidden by pages that paint an opaque background, and annotations (links, form fields) are always drawn above the page content, overlays included.

What is kept

Everything else in the document is kept: annotations, form fields and their values, bookmarks, named destinations, attachments, metadata, page labels and the structure tree of a tagged PDF, in which the stamp is marked as an artifact (a PDF/UA-1 document stamped this way still passes veraPDF, as does a PDF/A-2b one when the stamp has embedded fonts). The stamp page's own annotations, form fields and tags are not copied. The result is written out in full, not as an incremental update. A signed PDF is refused with `InvalidOperationException`, since stamping would break the signatures; set

`InvalidateSignatures = true` to stamp it anyway, or stamp before signing. Encrypted inputs open with `Password` (and `StampPassword` for the stamp); the result is not encrypted unless `Security` is set. Both inputs are handled as untrusted, like merge inputs: damaged files are repaired where possible, decoding is limited, and an unreadable file or one without pages throws `InvalidDataException`.

Text extraction

> **New in 1.9.0.**

`PdfText` extracts the text of any PDF, not only those PdfPecker made, in reading order: as one string, or page by page with every line and word and where it is on the page. It reads what the PDF draws, using PdfPecker's own PDF parser and font engines, so it needs no other package either.

```
// All the text, pages separated by a form feed (\f) as pdftotext does.
string text = PdfText.Extract(pdfBytes);
string report = PdfText.ExtractFile("report.pdf");

// Some pages of an encrypted PDF.
string summary = PdfText.Extract(pdfBytes, new PdfTextOptions { Password = "pw", Pages =
PdfPageRange.Parse("1-3") });

// Lines and words with their positions, such as for finding a total on an invoice.
foreach (PdfTextPage page in PdfText.ExtractPages(pdfBytes))
    foreach (PdfTextLine line in page.Lines)
        if (line.Text.StartsWith("Total"))
            Console.WriteLine($"Page {page.PageNumber}, {line.Y:F0} pt from the top:
{line.Words[^1].Text}");

PdfTextPage first = PdfText.ExtractPage(pdfBytes, 1);
```

`Extract`, `ExtractPages` and `ExtractPage` also take a `Stream`, and `ExtractPagesFromFile` reads the pages of a file.

Pages, lines and words

`PdfTextPage.Text` is the page's text, with lines separated by a line feed and paragraphs by an empty line. Its `Lines` go from top to bottom, and each `PdfTextLine` has its `Text`, its `Words` in reading order and a bounding box (`X`, `Y`, `Width`, `Height`). Each `PdfTextWord` has its `Text`, its box, its `FontSize` and its `FontName` (such as `Helvetica-Bold`). Positions are in points from the top-left corner of the page as it is displayed, after its rotation, with y growing downward, like the signature box and the stamp offsets. Boxes span the font's usual ascent and descent, so they are close to, not exactly, the inked area. Text that runs up or down the page gets its own lines, with their `Angle`.

What is read

The characters come from the fonts' ToUnicode maps, encodings and glyph names, for the standard fonts and for embedded TrueType, CFF, Type 1 and Type 3 fonts, and `/ActualText` in marked content stands for the glyphs it covers. Where a PDF draws no space between words but leaves a gap, the gap becomes a space; kerning and letter spacing do not. Ligatures such as `fi` come out as their letters unless `ExpandLigatures` is false. Right-to-left text such as Arabic and Hebrew comes out in logical order, and `PdfTextLine.RightToLeft` says when most of a line is. Invisible text, such as the recognised text layer of a scanned page, is included, and text hidden by optional content (a layer that is

off) is not. The values shown in form fields and the text of stamps are included unless `IncludeAnnotations` is false.

Pages are read in parallel (`MaxDegreeOfParallelism`, default one per processor), `MaxPages` (default 100,000) limits the work, and inputs are handled as untrusted, as with merging. A damaged file or page throws `InvalidDataException`, a wrong password `PdfPasswordException`.

Columns side by side are read across, line by line, not one column after the other, and table cells on one row make one line. Text drawn only as outlines or images has no characters to extract; a scanned page needs OCR first. See Limitations.

PDF to image

`PdfToImage` and `PdfRenderer` render PDF pages to PNG, JPEG or WebP (or raw RGBA pixels). This works for any PDF, not only PdfPecker's own output, and also uses no third-party packages: the PDF parser, font engines, JPEG and fax decoders, rasterizer and the PNG, JPEG and WebP encoders are all part of the library.

```
// Every page as PNG bytes (rendered in parallel), or one page (0-based index)
byte[][] pages = PdfToImage.Convert(pdfBytes, new ImageOptions { Dpi = 150 });
byte[] first = PdfToImage.ConvertPage(pdfBytes, 0);

// report.pdf -> out/report-1.png, out/report-2.png, ...
PdfToImage.ConvertFile("report.pdf", "out");

// JPEG or lossless WebP: report.pdf -> out/report-1.jpg, ... (or .webp)
PdfToImage.ConvertFile("report.pdf", "out", new ImageOptions { Format = PdfImageFormat.Jpeg,
Quality = 90 });
byte[][] webps = PdfToImage.Convert(pdfBytes, new ImageOptions { Format = PdfImageFormat.WebP
});

// Open once, render many pages; RenderedImage also exposes the RGBA pixels
var renderer = PdfRenderer.Open(pdfBytes, password: null);
RenderedImage image = renderer.RenderPage(2, new ImageOptions { Width = 800 });
image.SavePng("page3.png");
File.WriteAllBytes("page3.jpg", image.ToJpeg(quality: 85));
File.WriteAllBytes("page3.webp", image.ToWebP());
```

Output formats

Output formats (`ImageOptions.Format`, default `Png`; `RenderedImage.Encode()`/`Save()` use it too):

- **PNG**: lossless, RGB for opaque pages and RGBA otherwise.
- **JPEG**: baseline JFIF with the standard (IJG) quantisation tables scaled by `Quality` (1 to 100, default 85) and Huffman tables optimised per image. Below quality 90 colour is subsampled 4:2:0; from 90 it keeps full resolution (4:4:4), which keeps coloured text sharp. JPEG has no transparency, so a `Transparent` render is flattened onto `BackgroundColor` (white by default). At quality 90 a 150 DPI text page decodes at about 45 dB PSNR.
- **WebP**: always lossless (VP8L), alpha included, decoding to exactly the PNG's pixels and usually 50% to 70% smaller. Pages with up to 256 colours use a palette (packed 2, 4 or 8 pixels per byte when small); others use subtract-green and per-tile prediction, then LZ77, a colour cache and canonical prefix codes. Here `Quality` is the compression effort: higher spends more time searching for a smaller file and never changes a pixel. Lossy WebP is not written. WebP is limited to 16,384 pixels per side (JPEG to 65,535); larger images throw `NotSupportedException`.

Options

`ImageOptions` sets `Dpi` (default 150) or a target `Width` / `Height` in pixels, `BackgroundColor` or `Transparent`, `RenderAnnotations`, `Password`, `MaxDegreeOfParallelism`, `PngCompression`, `Format` and `Quality` (an `ArgumentOutOfRangeException` outside 1 to 100). A `PdfRenderer` is thread-safe and shares decoded fonts and images between pages. Encrypted files open with the user or owner password (`PdfPasswordException` otherwise); unreadable files throw `InvalidDataException`, as does a page that needs more than 20 million drawing operations (nested forms can multiply a few bytes into billions). Pass a cancellation token to bound the time a page may take.

What the renderer supports

- **Files**: cross-reference tables and streams, object streams, incremental updates, damaged files (rebuilt by scanning), RC4 and AES-128/256 encryption, all standard filters (Flate, LZW, ASCII85, ASCIIHex, RunLength, predictors).
- **Graphics**: paths, fills (non-zero and even-odd), strokes with joins, caps and dashes, clipping, anti-aliasing, all blend modes, constant alpha, soft masks, transparency groups, axial, radial, function and mesh shadings (types 1 to 7), tiling patterns, form XObjects, optional content (layers) and annotation appearances.
- **Text**: embedded TrueType, OpenType, CFF and Type 1 fonts, Type 3 fonts, CID fonts with CMaps, vertical text, all text rendering modes. The 14 standard fonts and non-embedded fonts are drawn with built-in Liberation and DejaVu-derived faces stretched to the document's widths.
- **Images**: every bit depth and colour space (Gray, RGB, CMYK, Lab, ICC-based, Indexed, Separation, DeviceN), baseline and progressive JPEG, JPEG 2000 (`JPXDecode`, including embedded alpha), JBIG2 (generic, text, symbol, refinement and halftone regions, with `JBIG2Globals`), CCITT fax, stencil masks, colour-key masks and soft masks.

For what the renderer does not support yet, see Limitations.

A wkhtmltopdf alternative for .NET

wkhtmltopdf was archived in January 2023 and gets no more fixes, security fixes included. It renders with a Qt WebKit engine from 2016, so flexbox, grid and most CSS written since then don't work, and it has to be deployed as a native binary of about 90 MB for each operating system. .NET wrappers such as DinkToPdf, Rotativa, NReco.PdfGenerator and WkHtmlToPdf-DotNet carry the same engine with them.

PdfPecker is a replacement written in plain C#. It is one NuGet package with no native files, it supports modern CSS, and it converts in your own process instead of starting a program for every PDF. It is free under the MIT license.

```
dotnet add package PdfPecker
```

Replace the wkhtmltopdf call

From the command line or a Process

```
// Before: one wkhtmltopdf process per document
var process = Process.Start("wkhtmltopdf", "--page-size A4 -T 20mm -B 20mm invoice.html
invoice.pdf");
process.WaitForExit();
```

```
// After
using PdfPecker;

HtmlToPdf.ConvertFile("invoice.html", "invoice.pdf", new PdfOptions
{
    PageSize = PageSize.A4,
    Margins = PageMargins.FromMillimeters(20, 10, 20, 10),
});
```

From DinkToPdf or WkHtmlToPdf-DotNet

```
// Before
var converter = new SynchronizedConverter(new PdfTools());
var document = new HtmlToPdfDocument
{
    GlobalSettings = { PaperSize = PaperKind.A4, Orientation = Orientation.Landscape },
    Objects = { new ObjectSettings { HtmlContent = html } },
};
byte[] pdf = converter.Convert(document);
```

```
// After
byte[] pdf = HtmlToPdf.Convert(html, new PdfOptions { PageSize = PageSize.A4.Landscape });
```

There is no converter object to keep alive and no native library to load: `HtmlToPdf` is static and thread-safe, so many requests can convert at the same time without the `SynchronizedConverter` lock.

From Rotativa in ASP.NET Core

Rotativa renders a Razor view and passes the HTML to wkhtmltopdf. Render the view to a string the way you already do for emails, or build the HTML from a template, and return the PDF:

```
app.MapGet("/invoice/{id}", (int id) =>
    Results.File(HtmlToPdf.Convert(RenderInvoice(id)), "application/pdf",
$"invoice-{id}.pdf"));
```

Options side by side

wkhtmltopdf	PdfPecker
`--page-size A4`, `-O Landscape`	`PageSize = PageSize.A4.Landscape`, or CSS `@page { size: A4 landscape }`
`-T`, `-R`, `-B`, `-L`	`Margins = PageMargins.FromMillimeters(top, right, bottom, left)`, or CSS `@page { margin: ... }`
`--footer-center "[page] of [topage]"`	`FooterText = "{page} of {pages}"`, or an `@page` margin box with `counter(page)` and `counter(pages)`
`--header-html header.html`	An element with `data-sticky-pdf-header` in the page (see Paged media)
`--title`	`Title`, or the document's `<title>`
`--outline`	On by default: bookmarks come from `h1` to `h6`
`toc`	Write the table of contents in your HTML with `target-counter()` and `leader()` for page numbers
`--print-media-type`	Always on: `@media print` rules apply
`--enable-local-file-access`	Files under `BaseDirectory` are read; `AllowFileAccess = true` allows others
`--zoom`, smart shrinking	`Scale`. PdfPecker does not shrink content to fit, so check page breaks after switching
`--javascript-delay`, `--run-script`	Not supported: render the content into the HTML before converting
`--enable-forms`	On by default: inputs become fillable fields (`FormFields = false` turns it off)

What changes in the output

- Pages can count differently. wkhtmltopdf shrinks content to fit the page by default, so a report that was 24 pages can become 29 at its real size. Set `Scale` if you need the old size back, but the real size is usually what the CSS meant.
- Flexbox, grid, `calc()`, CSS variables and other modern CSS now work, so workarounds written for the old WebKit, such as tables used for layout or `-webkit-box`, can be removed or left as they are.

- JavaScript is not run. Charts or text that scripts add to the page have to be in the HTML already, for example as SVG made on the server.
- Fonts come from files you register, not from the server's installed fonts, so the output is the same on every machine. See Fonts.

Speed and size

On the same 4-core Linux VM, a one-page invoice took 310 ms with wkhtmltopdf and 0.86 ms with PdfPecker, and a 30-page report 780 ms and 45 ms. Deployment drops from about 90 MB of native files to one 4 MB DLL. The full comparison is on the Performance page.

Next steps

- HTML to PDF in C#: a walk-through with templates, ASP.NET Core and fonts.
- HTML and CSS and Limitations: what your templates can use.
- Compare: PdfPecker next to wkhtmltopdf and the other .NET converters, feature by feature.

An HtmlRenderer.PdfSharp alternative

HtmlRenderer.PdfSharp is a free, managed HTML to PDF converter that runs on top of PDFsharp. Two things make people look for a replacement:

- **The licence of the engine underneath.** HtmlRenderer.PdfSharp 1.6.1 itself is BSD-licensed and still installs PDFsharp 6.2.4, the last MIT version. PDFsharp 6.5.0 (October 2026) and the 7.0 previews are AGPL-3.0 or a paid commercial licence. Updating PDFsharp, by hand or with a tool such as Dependabot, adding another package that needs a newer PDFsharp, or a future HtmlRenderer release built on it, brings in the AGPL, which requires you to publish your own application's source code unless you buy a licence. Staying on MIT means pinning PDFsharp at 6.2.4 and doing without its future fixes.
- **The CSS.** HtmlRenderer supports HTML 4 and CSS 2, so flexbox, grid, CSS variables, `calc()` and `@page` rules are ignored.

PdfPecker is also fully managed and runs in your own process, with no dependencies at all, under the MIT licence. It supports modern CSS and has PDF/A, PDF/UA, signatures, forms and merging built in.

```
dotnet add package PdfPecker
```

Then remove the `HtmlRenderer.PdfSharp` package, and `PDFsharp` too if nothing else in your project uses it.

Replace the HtmlRenderer call

```
// Before
using PdfSharp;
using TheArtOfDev.HtmlRenderer.PdfSharp;

var document = PdfGenerator.GeneratePdf(html, PageSize.A4);
document.Save("invoice.pdf");
```

```
// After
using PdfPecker;

File.WriteAllBytes("invoice.pdf", HtmlToPdf.Convert(html));
```

A4 is the default. With margins and a page footer:

```
// Before
var config = new PdfGenerateConfig { PageSize = PageSize.A4, MarginTop = 40, MarginBottom = 40
};
var document = PdfGenerator.GeneratePdf(html, config);
```

```
// After
byte[] pdf = HtmlToPdf.Convert(html, new PdfOptions
{
    Margins = PageMargins.FromMillimeters(14, 15, 14, 15),
    FooterText = "Page {page} of {pages}",
});
```

HtmlRenderer has no page numbers of its own; PdfPecker has `FooterText`, `@page` margin boxes and running headers and footers written in HTML (see Paged media).

Options side by side

HtmlRenderer.PdfSharp	PdfPecker
`PageSize`, `PageOrientation` in `PdfGenerateConfig`	`PageSize = PageSize.A4.Landscape`, or CSS `@page { size: A4 landscape }`
`MarginTop`, `MarginBottom`, `MarginLeft`, `MarginRight` (points)	`Margins = new PageMargins(top, right, bottom, left)` in points, or `PageMargins.FromMillimeters`
`cssData` argument	`UserStyleSheet`, or `<style>` and `<link>` in the HTML
`imageLoad` and `stylesheetLoad` events	`ResourceResolver`, or files under `BaseDirectory`
`PdfDocument` for further editing	PDF bytes; merge, stamp, sign and protect them with PdfPecker's own APIs
Fonts from PDFsharp's `GlobalFontSettings.FontResolver`	`options.Fonts.AddFile(...)` or `@font-face` (see Fonts)

What changes in the output

- Modern CSS now works. Layouts built with tables and floats to suit CSS 2 keep working, and you can move to flexbox and grid when you want.
- Backgrounds and colours that HtmlRenderer dropped, such as coloured table headers, are drawn.
- Long tables break across pages with their header row repeated, and rows are never cut in half.
- Text is laid out by PdfPecker's own engine, so line breaks and page breaks can fall in slightly different places. Check your longest templates after switching.

Speed and memory

On the same 4-core Linux VM, a one-page invoice took 6.3 ms with HtmlRenderer.PdfSharp and 0.86 ms with PdfPecker, a 30-page report 639 ms and 45 ms, and peak memory was 236 MB and 90 MB. See Performance.

When HtmlRenderer is still a fine choice

For simple HTML that already looks right, there is no urgent reason to move today: version 1.6.1 runs on the MIT PDFsharp 6.2.4. Pin that version so an update can't change the licence. PDFsharp itself remains a solid API for drawing PDFs from code. Compare shows the two side by side with the other .NET converters.

An IronPDF alternative

IronPDF converts HTML with an embedded Chrome engine and is sold per developer. PdfPecker converts the same kind of templates with its own engine written in C#: no Chrome binaries to deploy, no licence key, and an MIT licence that is free for commercial use. For invoices, payslips, reports and other HTML you control, the output can match IronPDF's to a fraction of a millimetre.

```
dotnet add package PdfPecker
```

Replace the IronPDF call

```
// Before
var renderer = new ChromePdfRenderer();
renderer.RenderingOptions.PaperSize = PdfPaperSize.A4;
renderer.RenderingOptions.MarginTop = 20;
renderer.RenderingOptions.MarginBottom = 20;
renderer.RenderingOptions.TextFooter = new TextHeaderFooter { CenterText = "{page} of {total-
pages}" };
using var document = renderer.RenderHtmlAsPdf(html);
byte[] pdf = document.BinaryData;
```

```
// After
using PdfPecker;

byte[] pdf = HtmlToPdf.Convert(html, new PdfOptions
{
    PageSize = PageSize.A4,
    Margins = PageMargins.FromMillimeters(20, 15, 20, 15),
    FooterText = "{page} of {pages}",
});
```

`HtmlToPdf` is static and thread-safe, so there is no renderer to create or dispose, and conversions can run in parallel. To write the PDF to a file or a response stream, pass the stream: `HtmlToPdf.Convert(html, stream)`.

Headers and footers in HTML

IronPDF's `HtmlHeaderFooter` takes a piece of HTML with `{page}` and `{total-pages}`. In PdfPecker the header or footer is an element of the page itself, marked with `data-sticky-pdf-header` or `data-sticky-pdf-footer`, and the same placeholders work:

```
<header data-sticky-pdf-header height="17mm"><img src="logo.png" height="40"> ACME
Ltd</header>
<main>...</main>
<footer data-sticky-pdf-footer height="12mm">Page {page} of {total-pages}</footer>
```

The element is drawn on every page and the content flows between them. See Paged media for `@page` margin boxes and running headers.

Matching IronPDF's output exactly

Chrome lays a page out at a whole number of CSS pixels and stretches it to the paper, so IronPDF's PDFs are a little larger than the CSS says. If you are switching an existing system and want every box and glyph where IronPDF put it, for example to compare old and new PDFs, use the same page geometry:

```
var options = new PdfOptions
{
    PageSize = new PageSize(594.96f, 841.92f),   // A4 as Chrome writes it
    Scale = 793.28f / 793f,
    TruncateBandsToWholePoints = true,           // header and footer heights rounded as
Chrome does
};
```

One user ran this on the 26 templates of their production app: 24 came out identical to IronPDF's output and the other two differed by less than 0.25 mm. For a new system, the defaults are fine.

Options side by side

IronPDF `RenderingOptions`	**PdfPecker** `PdfOptions`
`PaperSize`, `PaperOrientation`	`PageSize = PageSize.A4.Landscape`, or CSS `@page { size: A4 landscape }`
`MarginTop`, `MarginBottom`, `MarginLeft`, `MarginRight` (mm)	`Margins = PageMargins.FromMillimeters(top, right, bottom, left)`
`TextHeader`, `TextFooter`	`HeaderText`, `FooterText` with `{page}` and `{pages}`
`HtmlHeader`, `HtmlFooter`	`data-sticky-pdf-header` and `data-sticky-pdf-footer` elements
`CssMediaType = Print`	Always print: `@media print` rules apply
`PrintHtmlBackgrounds`	Always on, as in Chrome's print output
`Zoom`	`Scale`
`CreatePdfFormsFromHtml`	On by default (`FormFields = false` turns it off)
`Title` and `MetaData`	`Title`, `Author`, `Subject`, `Keywords` and `CustomProperties`
`EnableJavaScript`, `WaitFor`	Not supported: render the content into the HTML before converting
`RenderHtmlAsPdf(html, baseUrl)`	`BaseDirectory` for files, `ResourceResolver` for `http` URLs

PDF/A, PDF/UA, passwords, watermarks, digital signatures, ZUGFeRD and Factur-X, merging, stamping and PDF to image are built in too; each has a page in these docs.

Speed

On the same 26 production templates, IronPDF 2026.7.2 took 13.6 s for all of them one at a time and PdfPecker 17 ms. With 22 in parallel, IronPDF made 9.2 documents per second and PdfPecker 1,143. Details are on the Performance page.

When IronPDF is the better choice

IronPDF prints exactly what Chrome prints, runs JavaScript, converts live websites and comes with paid support and a large PDF editing API. If your pages build themselves with scripts, or you need output identical to a browser for arbitrary HTML, keep a Chrome-based converter. Compare lists what each converter can do.

PDF without Chrome: an alternative to PuppeteerSharp and Playwright

PuppeteerSharp and Playwright make PDFs by driving a real Chromium browser. That gives exactly what Chrome prints, but it means downloading and patching a browser of about 600 MB, keeping it running, and paying for a browser tab for every document. PdfPecker converts HTML to PDF in your own process with its own engine written in C#, so there is no browser to install and a one-page document takes about a millisecond. It is free under the MIT license.

```
dotnet add package PdfPecker
```

Replace the browser code

From PuppeteerSharp

```
// Before
await new BrowserFetcher().DownloadAsync();
await using var browser = await Puppeteer.LaunchAsync(new LaunchOptions { Headless = true });
await using var page = await browser.NewPageAsync();
await page.SetContentAsync(html);
byte[] pdf = await page.PdfDataAsync(new PdfOptions
{
    Format = PaperFormat.A4,
    PrintBackground = true,
    MarginOptions = new MarginOptions { Top = "20mm", Bottom = "20mm" },
});
```

```
// After
using PdfPecker;

byte[] pdf = HtmlToPdf.Convert(html, new PdfOptions
{
    PageSize = PageSize.A4,
    Margins = PageMargins.FromMillimeters(20, 15, 20, 15),
});
```

Both libraries have a class called `PdfOptions`, so in a project that still uses PuppeteerSharp elsewhere, write `PdfPecker.PdfOptions`.

From Playwright

```
// Before
using var playwright = await Playwright.CreateAsync();
await using var browser = await playwright.Chromium.LaunchAsync();
var page = await browser.NewPageAsync();
await page.SetContentAsync(html);
byte[] pdf = await page.PdfAsync(new PagePdfOptions { Format = "A4", PrintBackground = true
});
```

```
// After
byte[] pdf = await HtmlToPdf.ConvertAsync(html);
```

A4 is the default page size. `HtmlToPdf` is static and thread-safe, so there is no browser to share between requests and nothing to dispose or restart.

Headers and footers

Chrome's `headerTemplate` and `footerTemplate` use `<span class="pageNumber">` and `<span class="totalPages">`, and need `displayHeaderFooter`. In PdfPecker, use `FooterText = "{page} of {pages}"` for a simple line, or put the header or footer in the page itself:

```
<footer data-sticky-pdf-footer height="12mm">Page {page} of {total-pages}</footer>
```

Unlike Chrome's templates, these elements use the page's own stylesheets, fonts and images. `@page` margin boxes with `counter(page)` work too. See Paged media.

Options side by side

Chrome `page.pdf()` option	PdfPecker
`format`, `landscape`, `width`, `height`	`PageSize = PageSize.A4.Landscape`, or `new PageSize(width, height)` in points
`margin`	`Margins = PageMargins.FromMillimeters(top, right, bottom, left)`
`preferCSSPageSize`	Always on: CSS `@page` sets the size unless `IgnoreCssPageRules = true`
`printBackground`	Always on
`scale`	`Scale`
`displayHeaderFooter`, `headerTemplate`, `footerTemplate`	`HeaderText`, `FooterText`, or `data-sticky-pdf-header` and `data-sticky-pdf-footer` elements
`pageRanges`	Convert, then pick pages with the page tools
`outline`	On by default: bookmarks come from `h1` to `h6`
`tagged`	`Accessibility`, up to PDF/UA-2 (see Accessibility)

Chrome `page.pdf()` option	PdfPecker
`emulateMedia({ media: "print" })`	Always print: `@media print` rules apply
`waitForSelector`, `waitForNetworkIdle`	Not needed: nothing loads later, and JavaScript is not run

What changes

- JavaScript is not run, so content that scripts build in the browser has to be in the HTML before conversion. Templates rendered on the server (Razor, Scriban, string building) work as they are.
- Nothing is downloaded. Images and stylesheets come from files under `BaseDirectory`, `data:` URLs, or your own `ResourceResolver` for `http` URLs, so a template can't make the server fetch other addresses.
- PdfPecker supports a large part of modern CSS, including flexbox, grid, variables and `@page` rules, but not all of what Chrome does. Limitations lists the gaps.

Speed and memory

On the same 4-core Linux VM, a one-page invoice took 153 ms with PuppeteerSharp, 194 ms with Playwright and 0.86 ms with PdfPecker. Peak memory was 829 MB and 967 MB for the browser-based converters and 90 MB for PdfPecker, and in parallel PdfPecker made 818 documents per second against 8 to 11. The full table is on the Performance page.

When to keep Chrome

PuppeteerSharp and Playwright are free and print exactly what Chrome shows, so they remain the right tool for pages that build themselves with JavaScript, for live websites and for screenshots. Compare shows what each converter can do.

Changelog

What changed in each PdfPecker release on NuGet, newest first. The manual for any of these versions can be downloaded as a PDF with the version picker next to Download PDF; the source of every change is in the PdfPecker repository.

1.9.0

Released 16 October 2026.

- Chinese, Japanese and Korean line breaking with kinsoku rules, CSS `line-break` and `word-break: keep-all`.
- External signers: sign with a key in an HSM, smart card or cloud key vault, and `PdfSigner.SignAsync`.
- Signature verification and long-term validation (PAdES B-LT and B-LTA).
- Text extraction from any PDF, with the position of every line and word.
- Fill and flatten the forms of existing PDFs.
- Container queries and the `cq` units.
- Footnotes with `float: footnote`, numbered calls and a styled footnote area.
- Each page laid out at its own width when `:first`, `:left` or `:right` change it, and right-to-left page order.
- Blurred text shadows with a true Gaussian blur, shadows on underlines, and selectable text under `filter: blur()`.
- `aspect-ratio` in shrink-to-fit widths, with `min-` / `max-height` and on tables; `object-fit` for SVG images; and complex selectors inside `:is()`, `:where()`, `:not()` and `:has()`.

1.8.0

Released 16 October 2026.

- Variable fonts: weight, width, italic and optical size axes, `font-variation-settings`, kerning and CFF2 outlines (see Fonts).
- OpenType figures and features: `font-variant-numeric` (tabular, old-style, fractions and more) and `font-feature-settings` (see Fonts).
- `border-image`, CSS `filter`, `clip-path` on HTML elements, blend modes, 3D border styles, inset box shadows, `::first-letter` and `::first-line`, `@counter-style` and relative colours.
- Every SVG filter primitive, with primitive subregions, and text inside filtered SVG stays selectable.
- `<image>` in SVG, SVG watermarks, and BMP and TIFF images.
- `@import`, and `:first`, `:left` and `:right` page margins and sizes.
- The manual moved from the README to pdfpecker.net/docs, and the NuGet package links to it and has an icon.

1.7.0

Released 15 October 2026.

- Stamping: put a page of one PDF over or under the pages of another.
- Attachments: attach any file to a PDF and read attachments back.
- Automatic hyphenation with `hyphens: auto`.
- Add, change or remove passwords on existing PDFs.
- Extract, reorder, remove, rotate and split pages.
- JPEG and WebP output from `PdfToImage`.
- PDF/A-2u, 3u, 4 and 4f, and PDF/UA-2.
- Right-to-left text in SVG, and right-to-left ellipsis.

1.6.0

Released 15 October 2026.

- Right-to-left text with Arabic and Hebrew shaping.
- Named pages, running headers and footers from HTML, table of contents page numbers with `target-counter()` and `leader()`, and `conic-gradient()` (see Paged media).
- Much more of CSS3: modern colour syntax, `calc()` and the other math functions, `@layer`, nesting, `@supports`, `@property`, media queries, `:has()` and margin boxes.
- Sign into an existing empty signature field, merging keeps tagging and PDF/UA, and `PdfOptions.MissingCharacter` reports characters no font has.

1.5.1

Released 14 October 2026.

- Targets .NET 8 and 9 as well as .NET 10.
- Complete third-party licence notices, used as the NuGet package licence. 1.5.0 was never published, because nuget.org did not accept its licence expression.

1.4.0

Released 14 October 2026.

- Fillable PDF forms from HTML inputs, check boxes, radio buttons, selects and text areas.

1.3.0

Released 14 October 2026.

- Tagged PDF and PDF/UA-1 output.
- SVG masks, patterns, markers and filters, clipping in table cells, and double rounded borders.

1.2.0

Released 14 October 2026.

- Attach PDFs to Peppol and OIOUBL (UBL) e-invoices.
- Faster HTML to PDF, with half the memory allocations and decoded images reused.

1.1.1

Released 13 October 2026.

- The README shows its logo on nuget.org and has a NuGet badge. No changes to the library.

1.1.0

Released 13 October 2026.

- Factur-X and ZUGFeRD e-invoices.
- SVG, GIF and WebP images.
- Streaming PDF merging: stream and async overloads with one input in memory at a time.
- `calc()`, `min()`, `max()` and `clamp()`, `:nth-of-type`, `:is()`, `:where()`, `object-fit`, `text-overflow`, `outline`, `text-shadow` and `aspect-ratio`.
- Header and footer heights are rounded to whole points, as Chrome does.

1.0.0

Released 12 October 2026.

- The first release: HTML and Markdown to PDF and PDF to PNG with its own HTML parser, CSS engine, layout engine, PDF writer and renderer, and no dependencies outside the .NET base class library.

Limitations

> **Changed in 1.9.0:** Text and fonts, CSS, Reading and changing existing PDFs.

PdfPecker is built for fast conversion of HTML templates you control, not for rendering arbitrary web pages. These are the known gaps; most are scope choices for speed and size, and they are the first candidates for future work. Anything not listed in What is supported should be assumed missing.

By design

- No JavaScript. Scripts are never run, so content must be in the HTML before conversion (render it server-side first).
- No network access: `http:` / `https:` stylesheets and images are not downloaded. Supply them through your `ResourceResolver`, as files under `BaseDirectory`, or as `data:` URIs.
- `<canvas>`, `<iframe>`, `<video>`, `<audio>`, `<object>` and `<embed>` are not drawn, nor are file, range, colour and image inputs. Form fields have no JavaScript (no calculations, formats or submit actions), and a field inside a transformed box, a running header or footer, or a `position: fixed` box is drawn but not fillable.

Text and fonts Changed in 1.9.0

- Without registered fonts or `EmbedStandardFonts`, text uses the standard PDF fonts, so characters are limited to the Windows-1252 set (Western European languages); other characters are replaced with `?`. The bundled Liberation fonts add Central European, Greek and Cyrillic, but not CJK or other scripts: register fonts for those.
- Complex-script shaping covers Arabic and Hebrew (and other scripts written with Arabic letters, such as Persian and Urdu). Devanagari and other Indic scripts, Thai, Khmer, Myanmar and Syriac are not shaped, so their vowel signs and conjuncts come out wrong, and Thai text breaks only at spaces. There is no vertical text (`writing-mode`).
- Chinese, Japanese and Korean text breaks into lines, but `text-align: justify` stretches only spaces, so CJK lines without spaces are not justified (`text-justify` is not supported), and punctuation is not compressed at the start or end of a line.
- In Latin and other simple scripts, shaping is limited to kerning, standard ligatures and features that swap one glyph for another (see OpenType features); contextual alternates and other features that depend on neighbouring characters are not applied, and `small-caps` is synthesized from scaled capitals.
- Automatic hyphenation (`hyphens: auto`) has built-in patterns for English, Danish, German, Dutch and Norwegian only; other languages need their patterns in `PdfOptions.Hyphenation`. `hyphenate-limit-chars` is not supported.
- SVG `<text>` uses only the first value of position lists and has no `textPath`; in a text chunk with right-to-left text, `dx` on its `tspan`s is ignored.

CSS Changed in 1.9.0

- Container queries: no `scroll-state()` queries, `style()` works only on custom properties, and style containment (counter scoping) and the `contain` property are not implemented. A document with container queries takes about two to four times as long to lay out, and queries that keep changing the layout stop after six passes. In running headers and footers, `@page` margin boxes and SVG images, the `cq` units fall back to viewport units.
- Footnotes: `@footnote` inside `@page :first`, `:left` or `:right` is ignored, `footnote-policy: block` acts like `line`, and `footnote-display: compact` is always inline. A note split across pages does not carry its background or border onto the next page, and margins between block notes do not collapse.
- When `@page :first`, `:left` or `:right` give pages different widths, floats, inline-blocks, flex, grid and multi-column containers, and boxes with transforms or filters keep the width of the page they start on (Chrome re-sizes floats, inline-blocks and multi-column per page). A named page's `:first` is the document's first page, as in Chrome, not the first page of the named run.
- Selectors: a rule with an unsupported selector is skipped. A complex selector may have at most 64 compound parts and 16 levels of nesting, and `:has()` cannot contain `:has()`.
- Blurred text shadows are drawn as images at 144 dpi (down to 72 dpi for wide blurs, with the blur capped at about 200px), so shadowing long body text adds to the file size; the shadow of an underline is painted under the glyphs rather than between them and their own underline. `filter: blur()` and `drop-shadow()` draw the box as an image; its text stays selectable, and the colour filters keep everything as vectors.
- `text-overflow` applies only at the end of the line, as in the spec and Chrome. Percentage heights count as `auto` in block layout, also with `aspect-ratio`.
- `background-attachment: fixed` scrolls with the box, since paper has no scrolling viewport.

Backgrounds and transparency

- No `background-clip: text`; backgrounds of inline elements (`<span>`) show only their colour, not images or gradients.
- Blurred box shadows and repeating gradients with translucent stops are blended onto the background colour rather than truly see-through; SVG gradient stops with transparency are blended onto white.

Layout

- Grid: no subgrid.
- Multi-column: children are kept whole and balanced between columns; a single block is never split across columns.
- Overflow: a positioned child with a `z-index` inside a clipping box stays stacked within that box.

Images and SVG

- No AVIF images: decoding them needs an AV1 video decoder, which is not planned. Convert them to WebP, PNG or JPEG.
- SVG filters are applied in pixels at 144 dpi, so filtered shapes become an image (text keeps an invisible, selectable copy).

Reading and changing existing PDFs New in 1.9.0

- Text extraction reads columns that sit side by side across, line by line, rather than one column after the other, and it does not use the structure tree of a tagged PDF to find the reading order. Text drawn only as outlines or images is not extracted; there is no OCR.
- Form filling does not shape text, so values in Arabic, Hebrew, Devanagari and other scripts that need shaping are stored but not drawn correctly. XFA forms are not filled: their XFA data is removed and the AcroForm fields are filled instead. Field JavaScript (calculations, formats) is not run.
- Signature verification counts the updates made after a signature but does not analyse them against its permissions (no DocMDP or FieldMDP change analysis), so it cannot tell an allowed form fill from a changed page. Revocation is checked only when `RevocationMode` asks the platform to, not from the OCSP responses and CRLs stored in the PDF.

Markdown and the PDF renderer

- Markdown: no `[^1]` footnote syntax (use HTML with `float: footnote`), definition lists, math or front matter.
- `PdfRenderer` (PDF to image): no predefined CJK CMaps or non-embedded CJK fonts, and ICC profiles are not applied (ICC-based colours use their device equivalents). WebP output is lossless only (no lossy VP8), and JPEG output is baseline only (no progressive or arithmetic coding).

Performance

PdfPecker lays out documents in your own process, without a browser, so a typical one-page document takes about a millisecond. For memory and CPU per document, see System requirements.

Compared with other converters

The same three documents converted to A4 PDF by PdfPecker and six .NET alternatives, on one 4-core cloud VM (Intel Xeon @ 2.8 GHz, 33 MB L3, 16 GB, Linux, .NET 10.0.12). Times are medians after a 3-second warm-up, averaged over two runs; "first PDF" includes starting the browser where there is one. Parallel runs one conversion per core at once on the 5-page document. Peak memory is the proportional set size (PSS) of the whole process tree, so a browser's helper processes are included once and shared pages are not double-counted; .NET processes run with a 16 MB GC gen0 budget, as .NET would otherwise size it from the CPU cache.

Converter	Engine	To deploy	First PDF	Invoice, 1 page	Feature tour, 5 pages	Report, 30 pages	Parallel	Peak memory
PdfPecker	own, managed	4.0 MB DLL	0.40 s	**0.86 ms**	**3.5 ms**	**45 ms**	**818 docs/s**	**90 MB**
HtmlRenderer.PdfSharp 1.6.1	own, managed (CSS 2)	2.4 MB of DLLs	0.58 s	6.3 ms	21 ms	639 ms	72 docs/s	236 MB
iText pdfHTML 6.3.4	own, managed	11 MB of DLLs	1.44 s	34 ms	209 ms	3.4 s	9 docs/s	624 MB
wkhtmltopdf 0.12.6	Qt WebKit, native	about 90 MB with Qt	0.36 s	310 ms	356 ms	780 ms	11 docs/s	164 MB
PuppeteerSharp 25.13	Chromium 141	about 600 MB browser	1.43 s	153 ms	217 ms	792 ms	11 docs/s	829 MB
Microsoft.Playwright 1.63	Chromium 141	browser + 135 MB driver	1.36 s	194 ms	247 ms	726 ms	8 docs/s	967 MB
Syncfusion HTML to PDF 35.1.40	Chrome (Blink), bundled	about 545 MB with Chrome	2.83 s	264 ms	320 ms	1.1 s	2 docs/s	2,193 MB

How to read it:

- PdfPecker is 60 to 220 times faster than the Chromium-based converters on the 1 and 5-page documents and about 17 times faster on the 30-page report, and uses a tenth of their memory. Per-document overhead (a browser tab, a process start) dominates for the others on small documents.

- Output is not identical. Chromium is the reference for modern CSS and runs JavaScript, which PdfPecker does not. HtmlRenderer supports only HTML 4 and CSS 2 (it drops the invoice's table header colours), and wkhtmltopdf uses a 2016 WebKit, shrinks content to fit by default (24 pages for the report instead of 29) and was archived in 2023.
- iText pdfHTML lays out the documents itself, like PdfPecker, but its time grows faster than the page count: 209 ms for 5 pages and 3.4 s for the 30-page report, which is one long table.
- Syncfusion ran as a trial without a licence key, which adds a line of text to each page. Its `AdditionalDelay` (2 seconds of waiting for scripts by default) was set to 0, as the other Chromium converters don't wait either. With its default settings it scales content down, so the report took 13 pages instead of 29.
- PDF sizes for the 1 / 5 / 30-page documents: PdfPecker 12 / 27 / 126 KB, Chromium 72 / 139 / 171 KB, wkhtmltopdf 30 / 59 / 191 KB, HtmlRenderer 35 / 65 / 273 KB, iText 26 / 45 / 104 KB, Syncfusion 42 / 89 / 132 KB.
- Each converter was used the way its documentation suggests for many documents: PdfPecker and HtmlRenderer in process, wkhtmltopdf as one process per document, and PuppeteerSharp and Playwright with one shared browser and a new tab per document (Playwright in one shared browser context, tagged PDF output off for both). iText reuses one font provider per thread, as its knowledge base advises. Syncfusion keeps one converter per concurrent conversion with `ReuseBrowserProcess` on, as its support forum advises, so the parallel run starts four browsers.

IronPDF was not measured here. One user compared PdfPecker with **IronPDF 2026.7.2** (Chrome-based) on the 26 templates of their production app, on a 22-core development machine after warm-up: IronPDF took 13.6 s for all 26 documents one at a time and PdfPecker 17 ms (about 800 times faster), and with 22 in parallel IronPDF made 9.2 documents/s and PdfPecker 1,143 (about 125 times). 24 of the 26 documents came out identical; the other two differ by less than 0.25 mm.

To reproduce the table from a clone of the PdfPecker repository (needs wkhtmltopdf on `PATH` and a Chromium at `CHROMIUM_PATH`):

```
dotnet run --project bench/PdfPecker.Comparison -c Release
```

The comparison project is not part of `PdfPecker.sln`, so its third-party packages never reach the library or its tests. Your numbers will differ with hardware and documents.

Conversion speed

Measured with `bench/PdfPecker.Benchmarks` (in the PdfPecker repository) on a 4-core cloud VM (.NET 10, Release, after JIT warm-up):

Document	Pages	Median	Throughput
Invoice with a PNG logo and a styled table	1	0.55 ms	about 1,800 docs/s on one core
Feature tour (text, lists, tables, JPEG and PNG)	5	2.5 ms	about 1,700 docs/s on 4 cores
Dense report with 1,000 table rows	30	19.5 ms	about 1,500 pages/s on one core
Markdown guide (`guide.md`, headings, lists, code, a table) to PDF	2	0.9 ms	about 1,000 docs/s on one core

Parsing Markdown to HTML runs at roughly 20 to 45 MB/s, so it is a small part of a Markdown to PDF conversion.

The first conversion in a process also pays for JIT compilation (about 150 ms); later ones run at the speeds above.

Rendering PDF pages to images, same machine, one core per page (`PdfRenderer`, first render of each page of a freshly opened document, so fonts and images are decoded too):

Document	Render at 96 DPI	Render at 150 DPI	PNG encoding at 150 DPI
Invoice (1 page)	13 ms	19 ms	24 ms
Feature tour (5 pages)	7 ms per page	12 ms per page	26 ms per page
Large report (30 pages)	5 ms per page	10 ms per page	29 ms per page

Rendering the same page again (fonts and glyphs cached) takes about 5 ms at 150 DPI. Multi-page conversions render pages in parallel. Encoding one 150 DPI A4 page (1240 x 1754) takes about 15 to 60 ms as PNG, 20 to 70 ms as JPEG and 40 to 85 ms as WebP at the default quality, depending on content.

How it works

The source paths below are in the PdfPecker repository under `src/PdfPecker/`.

Markdown is first turned into HTML: `Markdown/BlockParser` finds the block structure line by line, following the CommonMark reference algorithm, `Markdown/InlineParser` handles emphasis (delimiter stack) and links (bracket stack), and `Markdown/HtmlRenderer` writes the HTML, which then takes the same path as any other document:

1. `Html/HtmlParser` builds a DOM in a single pass, applying the browser rules for implied end tags.
2. `Css/StyleResolver` cascades a user-agent stylesheet, presentational attributes, document stylesheets and inline styles. Rules are indexed by id, class and tag, and elements with identical inputs share one computed style. For a tagged PDF, `Pdf/StructureTree` then maps the elements to structure types and gives every text node, image and list marker a slot in that tree.
3. `Layout/LayoutEngine` lays out blocks, inline text (line breaking, justification), lists and tables (automatic column widths), paginating as it goes, and produces a flat list of draw operations. Inline content with right-to-left text gets its embedding levels from `Text/Bidi` (UAX #9), and each line is put in visual order after it is broken. `Fonts/ComplexShaper` shapes Arabic and Hebrew runs with the font's GSUB and GPOS tables (`Fonts/OpenTypeLayout`).
4. `Pdf/PdfDocumentWriter` writes PDF 1.4: content streams, fonts, image XObjects, link annotations, the bookmark outline and the cross-reference table. A tagged PDF's content is wrapped in marked-content sequences for its slots (or as artifacts), and the structure tree and its parent tree are written last. `Fonts/FontSubsetter` and `Fonts/CffSubsetter` cut embedded fonts down to the glyphs used (with their composite components and the CFF subroutines they call).

The renderer (`Rendering/`) works the other way round: `Parsing/PdfFile` reads the object graph, `PageRenderer` interprets content streams, `Fonts/` turns glyphs into outlines (cached as anti-aliased bitmaps), `Images/` decodes images, and `Graphics/Rasterizer` fills paths with exact area coverage before `PngEncoder`, `JpegEncoder` or `WebPEncoder` writes the result.

The third-party fonts, font data and code built into the library are listed, with their licenses, in THIRD-PARTY-NOTICES.md.